

Gelareto: Improving Latent Topology Prediction by Respecting Geometry

1 Abstract

Latent topology becomes more predictive when both representation learning and persistent-homology computation respect the data’s mixed geometry

Sequential data such as videos and time series are frequently modeled through latent representations, but these representations can miss higher-order structure in the evolving signal. In this work, we study whether topological summaries of latent trajectories can improve latent-space predictions. Given a sequence, we embed each state into a latent vector, form point clouds, compute persistence features, then combine them with the latent state as input to a downstream predictor. Across 9 video and time series datasets for 3 encoder families, adding topological features improves prediction in 24/27 settings and outperforms PCA, KPCA, manifold, and random-feature controls. Our experiments show that selected persistence descriptors frequently outperform matched linear, kernel, manifold, and random-feature controls, although the advantage depends on the persistence vectorization. When does latent topology improve sequential forecasting? We find that topological augmentation is broadly useful, but produces its strongest gains when the latent representation preserves metric geometry. We investigate how encoder geometry affects topological augmentation under latent-trajectory Vietoris-Rips persistence.

2 Introduction

Geometric latent representation topology (Gelareto)

In videos or dynamic systems, we often want to predict future observations, which is a computationally intensive task. Instead, many approaches operate on compressed latent representations learned by models such as autoencoders or convolutional neural networks (CNNs) [35, 34, 27]. However, these states might discard structural information that is useful for forecasting, especially when the sequence contains connected components, loops, or recurring geometric patterns. This motivates augmenting learned latent trajectories with topological descriptors computed from recent latent states. Although we focus on next-latent forecasting, the proposed representation is applicable to downstream tasks such as classification and anomaly detection [4, 22].

Existing latent forecasting methods typically predict future states from current or recent latent representations [7]. However, it remains unclear whether the topology of a latent trajectory contains complementary predictive information, or whether any improvement can instead be explained by generic nonlinear expansions or manifold geometry. We tackle this by computing persistent homology over sliding windows of latent states, rather than over individual frames, and compare it against linear (PCA), nonlinear (KPCA and random Fourier features), and manifold-learning (Laplacian Eigenmaps and Diffusion Maps) controls.

In this paper, we study whether persistent topology descriptors improve next-latent forecasting for sequential data. We encode each sequence state into a latent vector, compute Vietoris–Rips persistence on sliding windows of recent latent states, vectorize the resulting persistence diagrams, and use these descriptors as auxiliary inputs to an LSTM predictor. We evaluate 9 datasets across standard, geometry-regularized, and topology-regularized autoencoders. Perturbation and recoverability tests examine whether topology provides stable and complementary information beyond the latent representations, while PCA, KPCA, Laplacian eigenmaps, diffusion maps, and random Fourier feature controls test whether improvements arise from persistence topology rather than generic geometric or nonlinear representations. Overall, topology provides the largest gains when combined with GeoAE, suggesting that geometry regularization can strengthen topological augmentation.

2.1 Background: TDA

Topological Data Analysis (TDA) is a framework that uses algebraic topology to analyze data and extract topological descriptors [13], with persistent homology as its main technique [23, 40]. Persistent homology tracks the birth and death of topological features across a filtration [23], producing summaries such as persistence barcodes or persistence

diagrams [14]. Figure 1 illustrates the Vietoris-Rips (VR) filtration process. These summaries can then be vectorized and used in statistical analysis or machine learning [57].

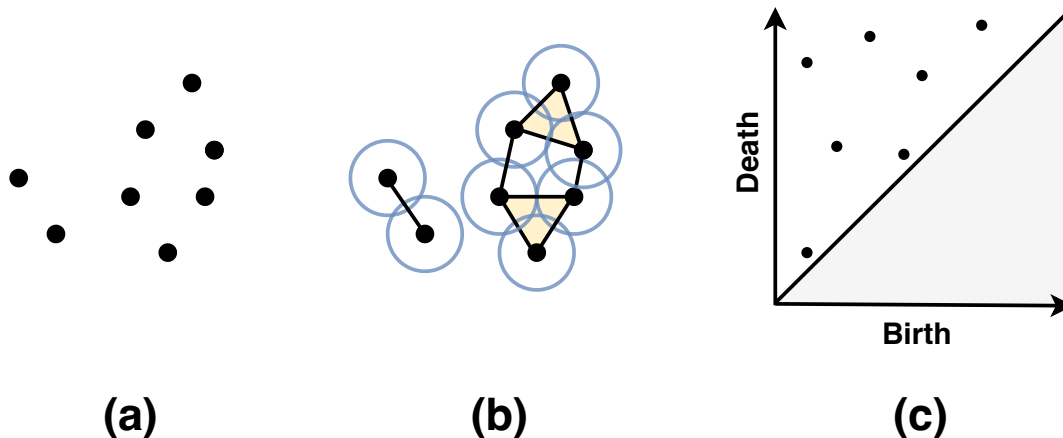


Figure 1: The Vietoris-Rips persistence. (a) Input point cloud. (b) Growing balls around each point define the Vietoris-Rips filtration: overlapping pairs include edges, and higher-order overlaps induce filled simplices. (c) The resulting persistence diagram records when topological features appear and disappear across the filtration.

The main homology dimensions used in this paper are H_0 , which describes connected components, and H_1 , which describes loop-like structures. We also use many vectorizations of persistence diagrams such as Betti curves, persistence images, persistence landscapes, and PersLay-style features. Betti curves count the number of active topological features across filtration values, while persistence images and landscapes retain more information about birth-death locations in the persistence diagram [23].

Persistence-based methods are used in many fields, including material science to detect cavities [33], hurricane analysis [51], and cosmology to describe Cosmic Web structures [41]. Persistence homology is also robust under small input noise [19], making it useful for applications such as denoising [32], anomaly detection [56], image segmentation [18], and feature extraction for machine learning [9].

TDA is applicable to many modalities, such as time series [30], videos [50], images [1], point clouds [13], and graphs [2]. For time series, a common approach is to construct a point cloud using time-delay embedding [47], then applying a distance-based filtration such as VR [26]. In contrast, our main pipeline constructs point clouds from sliding windows of learned latent states and computes VR persistence on them, avoiding a separate delay-embedding step.

2.2 Related Work

Persistent homology is extensively used as a feature representation, regularizer, and representation-learning objective in machine learning. Many works use persistent homology as a feature extractor for time series learning. For example, [37] compute persistence diagram statistics for classification, [52] uses Betti sequences as persistence-based features, and [30] combine time-delay embeddings with persistence vectorizations such as Betti curves and persistence landscapes. In contrast, we treat topology as complementary structure, augmenting learned latent representations rather than replacing them.

PersLay [15] is a neural layer that maps persistence diagrams to fixed-length vectors for downstream machine learning models. Unlike PersLay, we use persistence vectorizations as supplementary features along the learned latent state, rather than the primary representation.

TopoCL incorporates topology into representation learning via contrastive learning objective [31]. On the other hand, we use explicit topological descriptors to augment learned latent representations for downstream prediction.

The JEPA family predicts latent representations rather than raw observations. I-JEPA [5] predicts latent targets for masked regions within images, while V-JEPA [6] extends this idea to masked regions in future video frames. These models pretrain general-purpose encoders on large external datasets, whereas we focus on lightweight dataset-specific encoders and study whether persistent topology features improve next latent forecasting on learned representations.

We use V-JEPA only as an appendix robustness check (see Section B) to evaluate whether topology can also improve frozen pretrained representations.

3 Methodology

3.1 New method

Our method is shown in Fig. 2. Put simply, we use the preliminary curvature diagnostic of [25] to identify candidate latent geometries, then encode input states using a geometry-preserving autoencoder, then construct a Vietoris-Rips filtration on the latent windows using the selected manifold metric. We then concatenate the resulting persistence features with the current latent state to predict the next latent state.

In more detailed terms:

1. **Estimate Curvature:** Given a dataset $X = \{x_t\}_{t=1}^T$ of T states, apply the triangle-curvature diagnostic, inspired by [25], to identify candidate Euclidean ($\mathbb{E}$), spherical ($\mathbb{S}$), and hyperbolic ($\mathbb{H}$) components. The diagnostic narrows the search space of potential signatures, while the dimensions of the components are selected using validation performance.
2. **Define Manifold:** We define the product manifold

$$\mathcal{M} = \mathbb{E}^{d_E} \times \mathbb{S}^{d_S} \times \mathbb{H}^{d_H}, \quad d_E + d_S + d_H = d_z,$$

and build a convolutional autoencoder whose latent representations lie on $\mathcal{M}$. Following the standard product metric used in mixed-curvature representation learning [25], the distance between two latent points is

$$d_{\mathcal{M}}(z_i, z_j) = \sqrt{\|z_i^E - z_j^E\|_2^2 + d_{\mathbb{S}}(z_i^S, z_j^S)^2 + d_{\mathbb{H}}(z_i^H, z_j^H)^2}.$$

Following the constant-curvature formulation of [25], our method fixes the sectional curvatures to $K_{\mathbb{E}} = 0$, $K_{\mathbb{S}} = +1$, and $K_{\mathbb{H}} = -1$:

$$d_{\mathbb{S}}(u, v) = \arccos(\langle u, v \rangle), \quad d_{\mathbb{H}}(u, v) = \operatorname{arcosh}(-\langle u, v \rangle_L),$$

where the Lorentzian inner product is $\langle u, v \rangle_L = -u_0 v_0 + \sum_{k=1}^{d_H} u_k v_k$.

3. **Encode Observations:** Encode each observation x_t as $z_t = f_{\theta}(x_t) \in \mathcal{M}$, and reconstruct it as $\hat{x}_t = g_{\phi}(z_t)$. The autoencoder is trained using the reconstruction loss

$$\mathcal{L}_{\text{recon}} = \text{MSE}(X, \hat{X})$$

and the geometry-preservation loss

$$\mathcal{L}_{\text{geo}} = \frac{1}{N^2} \left\| \hat{D}_X - \hat{D}_{\mathcal{M}} \right\|_F^2,$$

where N is the number of observations used to compute the pairwise distance matrices, $(D_X)_{ij} = d_{\mathbb{E}}(x_i, x_j)$, $(D_{\mathcal{M}})_{ij} = d_{\mathcal{M}}(z_i, z_j)$, and the hats denote normalization by the mean pairwise distance. The objective is

$$\mathcal{L}_{\text{AE}} = \mathcal{L}_{\text{recon}} + \lambda_{\text{geo}} \mathcal{L}_{\text{geo}}.$$

λ_{geo} is the weighting coefficient of $\mathcal{L}_{\text{geo}}$

4. **Compute Persistence:** At time t , collect the most recent w latent points into the temporal point cloud

$$Z_t = \{z_{t-w+1}, \dots, z_t\}.$$

We construct a Vietoris-Rips filtration on Z_t using $d_{\mathcal{M}}(z_i, z_j)$ as its distance function. Persistent-homology features are then computed from this filtration to obtain $B_t^{\mathcal{M}}$.

5. **Concatenate Features:** Form the geometric-topological representation $h_t = [z_t, B_t^{\mathcal{M}}]$
6. **Forecast:** Feed h_t to the predictor to estimate the next latent state $\hat{z}_{t+1} = p_{\psi}(h_t)$, where ψ denotes the predictor parameters. The predicted latent state may then be decoded as

$$\hat{x}_{t+1} = g_{\phi}(\hat{z}_{t+1}).$$

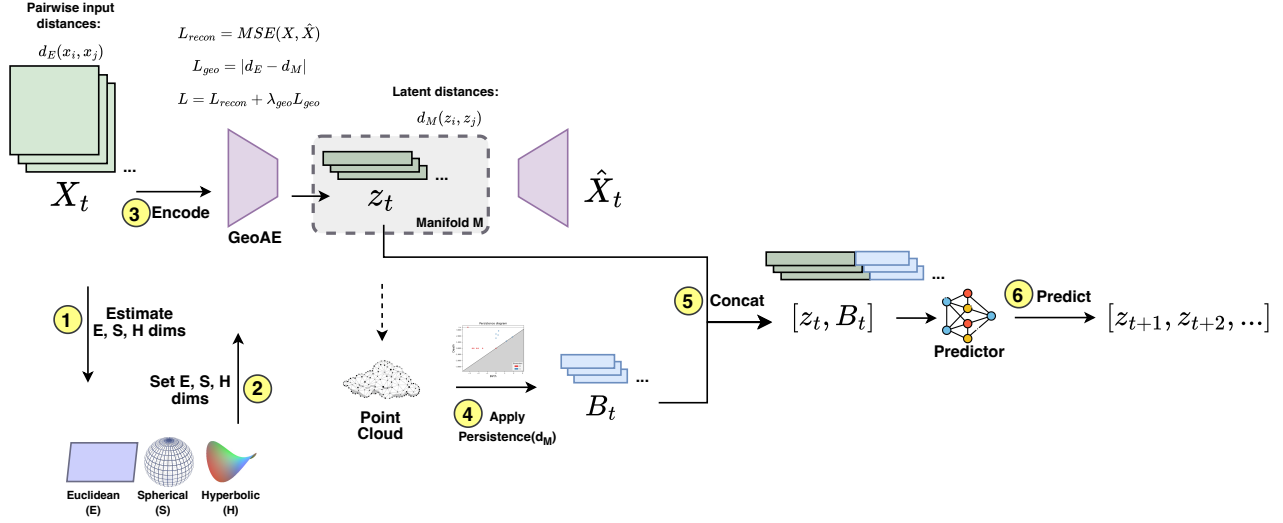


Figure 2: Diagram of the proposed method. Each input state x_t is encoded as a manifold latent representation z_t , then persistence features are computed over a sliding latent window using the manifold metric, concatenated with z_t , and used to predict the next latent state

3.2 Overview

Our main method, shown in Fig. 2, embeds each input frame or state X_t at time step t into a latent vector z_t . For each t , we form a latent trajectory sliding window point cloud

$$P_t = \{z_{t-w+1}, \dots, z_t\},$$

where w is the sliding window length, and compute the VR persistence on P_t . The resulting topological descriptor B_t , such as H_0 or H_1 Betti curves or other persistence-vectorized features, is used as auxiliary information for next-latent state prediction. To combine z_t and B_t , we use early-fusion variants inspired by multimodal learning [8]. Before fusion, we standardize z_t and B_t separately using training-set statistics to prevent scale mismatch between the two feature groups. In direct concatenation, the predictor receives $[z_t, B_t]$. In learned fusion, a small Multi-Layer Perceptron (MLP) maps $[z_t, B_t]$ to a fused representation $z_{\text{topo},t}$, which is then passed to the predictor.

Because Vietoris-Rips persistence is permutation-invariant, these descriptors depend on pairwise distances within P_t , but not on the temporal order of the latent states. So, the topological branch summarizes the geometry of the recent latent trajectory, while temporal order is modeled by the downstream LSTM predictor.

We emphasize H_1 descriptors in most controls because they capture loop-like structure, which is relevant for ring-shaped objects, holes, and recurring trajectories. Our stability diagnostic also shows that H_1 diagrams are less sensitive to input perturbations than H_0 diagrams in our latent point clouds, with $d_B(H_1)/d_H < d_B(H_0)/d_H$ across all dataset-encoder pairs (Appendix Table 11). We still include H_0 and $H_0 + H_1$ ablations to test this choice, rather than assume it.

We include a lightweight PersLay-style vectorization to obtain supplementary topological features for next latent prediction, rather than as the core prediction architecture.

We use a CNN encoder given its effectiveness in encoding sequential data and describing each frame’s contents [34], and a Long Short-Term Memory (LSTM) predictor for next-latent forecasting, as it is effective in learning time-based changes and handling sequential predictions [28]. For each dataset and seed, the encoder is trained then frozen; all topology and control modes are evaluated using the same frozen latent trajectories.

3.3 Autoencoder Variants

Our geometry-regularized autoencoder (GeoAE) is a convolutional autoencoder trained with reconstruction loss, plus a pairwise-distance preserving penalty. This penalty encourages normalized pairwise distances between latent codes to match normalized pairwise distances between the corresponding flattened input frames (ie, it penalizes distortions of the data’s metric structure within each mini-batch). This resembles the work of [17], though they use a continuous k-Nearest Neighbors graph objective, while we use a simpler mini-batch pairwise-distance penalty. The loss $\mathcal{L}_{\text{GeoAE}}$ is

given by:

$$\mathcal{L}_{\text{GeoAE}} = \mathcal{L}_{\text{recon}}(x, \hat{x}) + \lambda \left\| \frac{D_X}{\overline{D_X}} - \frac{D_Z}{\overline{D_Z}} \right\|_F^2,$$

D_X is the pairwise Euclidean distance matrix between flattened input frames, D_Z is the pairwise Euclidean distance matrix between their latent vectors, and $\overline{D_X}$ and $\overline{D_Z}$ denote the mean entries of the respective corresponding distance matrices. The pairwise distance d between two vectors x_1 and x_2 is

$$d(x_1, x_2) = \sqrt{\sum_k (x_{1,k} - x_{2,k})^2}$$

The topology-regularized autoencoder (TopoAE) adds an H_0 persistence penalty between the input mini-batch X and its latent representation Z , following [39]:

$$\mathcal{L}_{\text{TopoAE}} = \mathcal{L}_{\text{recon}}(x, \hat{x}) + \lambda \mathcal{L}_{H_0}(X, Z)$$

Here, X denotes the mini-batch of flattened input frames, treated as a point cloud in pixel space, and Z denotes the corresponding latent codes. The term $\mathcal{L}_{H_0}(X, Z)$ compares the finite H_0 Vietoris-Rips persistence pairs of X and Z by matching the selected Euclidean pairwise distances at which connected components merge.

3.4 Metric Distortion

Given a window of input frames $X_t = \{x_{t-w+1}, \dots, x_t\}$ and their corresponding latent representations $Z_t = \{z_{t-w+1}, \dots, z_t\}$, we define the pairwise distance matrices as

$$(D_X)_{ij} = \|\text{vec}(x_i) - \text{vec}(x_j)\|_2, \quad (D_Z)_{ij} = \|z_i - z_j\|_2,$$

where $\text{vec}(x_i)$ denotes the flattened input frame and $\|\cdot\|_2$ is the Euclidean norm.

Because the input and latent spaces may have different global scales, we normalize each distance matrix by its mean off-diagonal distance:

$$\overline{D_X} = \frac{1}{w(w-1)} \sum_{i \neq j} (D_X)_{ij}, \quad \overline{D_Z} = \frac{1}{w(w-1)} \sum_{i \neq j} (D_Z)_{ij}.$$

We then measure scale-normalized metric distortion as

$$\delta_{\text{metric}}(X_t, Z_t) = \frac{1}{w} \left\| \frac{D_X}{\overline{D_X}} - \frac{D_Z}{\overline{D_Z}} \right\|_F,$$

where w is the number of states in the window and $\|\cdot\|_F$ is the Frobenius norm [24]. Lower δ_{metric} indicates that the latent representation better preserves the pairwise geometry of the input trajectory.

3.5 Persistence Fidelity

Vietoris-Rips persistence depends only on pairwise distances. Therefore, a latent representation that preserves the input distance matrix should also preserve its persistence structure. This intuition is formalized by the stability theory of persistent homology [16]. Suppose that the normalized pairwise distances satisfy

$$\max_{i,j} \left| \frac{(D_X)_{ij}}{\overline{D_X}} - \frac{(D_Z)_{ij}}{\overline{D_Z}} \right| \leq \varepsilon.$$

Standard stability results for Vietoris-Rips persistence then imply [16]

$$d_B(\text{Dgm}_k(X_t), \text{Dgm}_k(Z_t)) \leq C\varepsilon,$$

where d_B is the bottleneck distance, k is the homology dimension, and C depends on the filtration convention. We therefore define persistence-fidelity error as

$$\delta_{\text{PH},k}(X_t, Z_t) = d_B(\text{Dgm}_k(X_t), \text{Dgm}_k(Z_t)).$$

Lower $\delta_{\text{PH},k}$ indicates that persistence computed in latent space more faithfully represents the persistence of the input trajectory.

Finally, we define the relative forecasting gain from topology as

$$G_{\text{topo}} = \frac{\text{MSE}(z) - \text{MSE}(z + B)}{\text{MSE}(z)}.$$

We empirically test whether lower metric distortion corresponds to lower persistence-fidelity error and whether both are associated with larger forecasting gains.

3.6 Topological Augmentation

Our topology descriptor B_t summarizes a sliding window of recent latent states and is provided as auxiliary input to the sequential predictor. The following proposition characterizes when this augmentation can reduce optimal prediction risk. Let $\mathcal{R}^*(U)$ denote the minimum expected mean-squared prediction error achievable using information U .

Proposition 1. *Let H_t denote any representation of the latent history available to a predictor, and let B_t denote the corresponding persistent homology descriptor. Assume that the future target Y_t has finite second moment. Then the optimal prediction risks satisfy*

$$\mathcal{R}^*(H_t) - \mathcal{R}^*(H_t, B_t) = \mathbb{E} \left[\|\mathbb{E}[Y_t | H_t, B_t] - \mathbb{E}[Y_t | H_t]\|_2^2 \right] \geq 0, \quad (1)$$

where Y_t is the future target.

Proof. Let $m_H = \mathbb{E}[Y_t | H_t]$ and $m_{HB} = \mathbb{E}[Y_t | H_t, B_t]$. Since

$$Y_t - m_H = (Y_t - m_{HB}) + (m_{HB} - m_H),$$

taking squared norms and expectations yields

$$\begin{aligned} \mathbb{E}\|Y_t - m_H\|^2 &= \mathbb{E}\|Y_t - m_{HB}\|^2 + \mathbb{E}\|m_{HB} - m_H\|^2 \\ &\quad + 2\mathbb{E}[\langle Y_t - m_{HB}, m_{HB} - m_H \rangle]. \end{aligned}$$

The final term equals zero by the Pythagorean orthogonality property of conditional expectation [54]. Therefore,

$$\mathbb{E}\|Y_t - m_H\|^2 = \mathbb{E}\|Y_t - m_{HB}\|^2 + \mathbb{E}\|m_{HB} - m_H\|^2.$$

Rearranging proves the result. $\square$

Proposition 1 shows that adding topology cannot increase the optimal prediction error under squared loss. The improvement is positive when (B_t) changes the expected future target after accounting for the latent history represented by (H_t) , and zero otherwise. This population-level result concerns the optimal predictor and does not guarantee improvement for a finite model trained on finite data.

3.7 Baselines

For non-topological controls, we replace the persistence descriptor with Principal Component Analysis (PCA) [49] or Kernel PCA (KPCA) [44] summaries computed from the same sliding-window latent point cloud. PCA tests whether gains come from adding a linear summary of latent-window variance, while KPCA provides a stronger nonlinear geometric control that captures curved latent structure without using persistence.

As non-topological manifold geometric controls, we compare against graph-spectral summaries computed from the same sliding-window latent point clouds used for persistence. For each window $P_t = \{z_{t-w+1}, \dots, z_t\}$, we build a weighted radial basis function (RBF) graph with affinities $W_{ij} = \exp(-\|z_i - z_j\|_2^2 / \sigma^2)$, where σ is chosen using the median pairwise-distance heuristic. We evaluate two manifold learning descriptors: 1) *Laplacian Eigenmaps*, using the smallest eigenvalues of the normalized graph Laplacian [11, 53], and 2) *Diffusion maps*, using the leading eigenvalues of the normalized diffusion operator [20], which summarize multiscale connectivity and diffusion geometry of the latent window.

As a dimension-matched nonlinear control, we compute random Fourier features (RFFs) from each standardized latent window [42], using the same fixed random map for training and testing. This tests whether generic nonlinear features explain the gains. Together, PCA/KPCA test linear and kernel variation, Laplacian and diffusion maps test manifold geometry, and RFFs test generic nonlinear expansion. These controls test whether improvements come from persistence-specific topology rather than generic linear, nonlinear, geometric, or manifold summaries of the latent window.

4 Experiments

4.1 Datasets

We run our methods on many sequential videos and time series datasets, both real and synthetic. The datasets, outlined in Table 1, are as follows:

Table 1: Datasets used in the forecasting experiments. Sequence counts report the effective train/test clips used in the pipeline.

Dataset	Type	Modality	Sequence length	Frame size	Sequences	Source
Bouncing disks	Synthetic	Video	50	64×64	512/128	-
Bouncing rings	Synthetic	Video	50	64×64	512/128	-
Orbiting disks	Synthetic	Video	50	64×64	512/128	-
Orbiting rings	Synthetic	Video	50	64×64	512/128	-
Moving MNIST	Real	Video	20	64×64	1000/200	[link] [46]
HeLa	Real	Video	50	64×64	14/28	[link] [38]
Glioblastoma	Real	Video	50	64×64	128/64	[link] [38]
Lorenz-96	Synthetic	Time series	100	64×64	512/128	-
ElectricDevices	Real	Time series	96	32×32	1024/512	[link] [21]

- Bouncing disks/rings: synthetic videos with multiple filled disks, or hollow rings, undergoing irregular bouncing motion.
- Orbiting disks/rings: synthetic videos with filled disks, or hollow rings, following smoother periodic orbital motion.
- Lorenz-96: chaotic multivariate time series generated from the Lorenz-96 dynamical system [36] with 40 variables and forcing $F = 8$, rendered as 64×64 heatmap-like frame sequences.
- Moving MNIST [46]: videos of two moving handwritten digits.
- ElectricDevices [21]: real electric-device usage time series rendered as frame sequences.
- Glioblastoma and HeLa [38]: real microscopy videos from the Cell Tracking Challenge.

Some dataset frames are shown in Fig. 3.

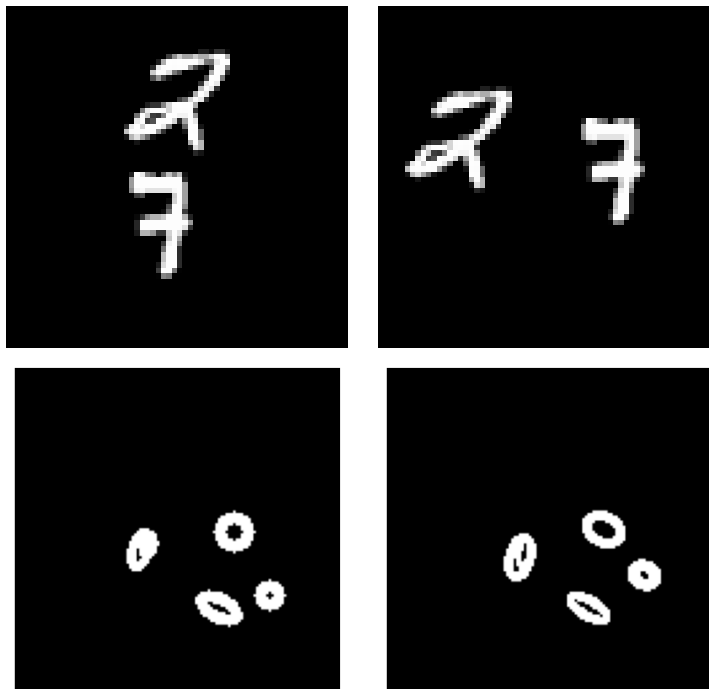


Figure 3: Example frames from the Moving MNIST (top) and bouncing balls (bottom) datasets.

4.2 Forecasting Results

Experimental details are given in Section C. We predict the future frames’ latent representation and evaluate them using the Mean Square Error (MSE) and coefficient of determination R^2 . Results are averaged over 5 seeds with

Table 2: Main latent-forecasting results after selecting predictor hyperparameters on z -only validation MSE. Each dataset is evaluated with a standard AE, a geometry-regularized AE (GeoAE), and a topology-regularized AE (TopoAE). “Direct” denotes fixed topology descriptors appended to the latent state, while “Fusion” learns $z_{\text{topo}} = \text{MLP}([z, B])$ jointly with the predictor. Bold marks the best entry within each encoder row; * marks the best result for the dataset.

Dataset	Encoder	z only	Best direct z +topology	Best fusion
Bouncing disks	AE	0.3429 $\pm$ 0.0266	0.3402$\pm$0.0256	0.3639 $\pm$ 0.0297
	GeoAE	0.0013 $\pm$ 0.0003	0.0010$\pm$0.0002*	0.0019 $\pm$ 0.0007
	TopoAE	0.2158 $\pm$ 0.0099	0.2070$\pm$0.0083	0.2259 $\pm$ 0.0075
Bouncing rings	AE	0.2972 $\pm$ 0.0474	0.2955$\pm$0.0495	0.3112 $\pm$ 0.0561
	GeoAE	0.0013 $\pm$ 0.0001	0.0011$\pm$0.0002*	0.0022 $\pm$ 0.0002
	TopoAE	0.2104 $\pm$ 0.0120	0.2024$\pm$0.0103	0.2160 $\pm$ 0.0105
Orbiting disks	AE	0.0052 $\pm$ 0.0018	0.0047$\pm$0.0016	0.0068 $\pm$ 0.0027
	GeoAE	0.0017 $\pm$ 0.0001	0.0011$\pm$0.0001*	0.0015 $\pm$ 0.0001
	TopoAE	0.2281 $\pm$ 0.0124	0.2083$\pm$0.0076	0.2761 $\pm$ 0.0162
Orbiting rings	AE	0.0045 $\pm$ 0.0014	0.0040$\pm$0.0013	0.0054 $\pm$ 0.0020
	GeoAE	0.0017 $\pm$ 0.0001	0.0010$\pm$0.0001*	0.0016 $\pm$ 0.0002
	TopoAE	0.2321 $\pm$ 0.0144	0.2254$\pm$0.0080	0.2617 $\pm$ 0.0202
Moving MNIST	AE	0.0048 $\pm$ 0.0019	0.0046$\pm$0.0018	0.0080 $\pm$ 0.0030
	GeoAE	0.0009$\pm$0.0001*	0.0009 $\pm$ 0.0001	0.0023 $\pm$ 0.0005
	TopoAE	0.2868 $\pm$ 0.0056	0.2748$\pm$0.0061	0.3037 $\pm$ 0.0119
Lorenz-96	AE	0.1536 $\pm$ 0.0057	0.1420$\pm$0.0119	0.1998 $\pm$ 0.0223
	GeoAE	0.0012$\pm$0.0003*	0.0012 $\pm$ 0.0001	0.0013 $\pm$ 0.0003
	TopoAE	0.2726$\pm$0.0298	0.2745 $\pm$ 0.0394	0.3264 $\pm$ 0.0258
ElectricDevices	AE	0.0139 $\pm$ 0.0072	0.0102$\pm$0.0031	0.0167 $\pm$ 0.0094
	GeoAE	0.0014 $\pm$ 0.0001	0.0011$\pm$0.0002*	0.0017 $\pm$ 0.0002
	TopoAE	0.0615 $\pm$ 0.0047	0.0492$\pm$0.0025	0.1120 $\pm$ 0.0063
HeLa	AE	0.0437 $\pm$ 0.0107	0.0383$\pm$0.0083	0.0496 $\pm$ 0.0081
	GeoAE	0.0072 $\pm$ 0.0032	0.0071$\pm$0.0033*	0.0133 $\pm$ 0.0054
	TopoAE	0.0587 $\pm$ 0.0365	0.0550$\pm$0.0324	0.0644 $\pm$ 0.0351
Glioblastoma	AE	0.0027 $\pm$ 0.0010	0.0021$\pm$0.0006	0.0071 $\pm$ 0.0016
	GeoAE	0.0011 $\pm$ 0.0001	0.0008$\pm$0.0001*	0.0063 $\pm$ 0.0035
	TopoAE	0.0040 $\pm$ 0.0013	0.0033$\pm$0.0008	0.0078 $\pm$ 0.0015

prediction horizon 5 and latent window size 15; see Appendix Fig. 7 for sensitivity across horizons. Lower MSE and higher R^2 are better.

Table 2 reports the main latent forecasting results. Consistent with Proposition 1, direct topology augmentation improves over the z -only baseline in 24/27 dataset–encoder settings, suggesting that persistence descriptors provide complementary predictive structure beyond the compressed latent representation. GeoAE gives the lowest MSE on all nine datasets, while direct augmentation consistently outperforms learned fusion. The compact results are reported in Table 15.

Table 3 compares the original H_1 descriptor with the best of four corrupted controls: zeroing, shuffling, additive noise, and temporal shifting. Because the corrupted descriptors can remain competitive with both z -only and the original H_1 descriptor, this experiment is a diagnostic of feature augmentation and alignment rather than evidence that the exact persistence values alone explain the gains.

Table 4 compares topology with PCA, KPCA, Laplacian eigenmaps, diffusion maps, and dimension-matched RFF controls computed from the same latent windows. Topology outperforms these controls in most dataset–encoder settings, suggesting that the gains are not explained solely by linear variation, manifold geometry, or nonlinear expansion.

Figure 4 visualizes the improvements in MSE under various encoders and feature combinations, of Tables 2 and 4. Topology improves over the z -only baseline in 24/27 dataset–encoder settings and outperforms the Laplacian, diffusion, and RFF controls in 26/27, 23/27, and 26/27 settings, respectively. GeoAE gives the lowest topology-augmented MSE on all nine datasets.

Appendix Table 14 reports results using frozen V-JEPA2 features. Although frozen V-JEPA2 performs worse than the dataset-specific encoders in this setting, topology consistently improves over z -only forecasting on all but one dataset.

Table 3: H_1 perturbation controls after z-only hyperparameter selection. “Best corrupt” is the best of zero, shuffle, noise, and temporal-shift controls. Corrupted controls can remain competitive, so this table is a diagnostic control rather than evidence that exact H_1 semantics alone explain the gains. Lower MSE is better.

Dataset	Encoder	z	$z + H_1$	$z + \text{best corrupt } H_1$
Bouncing disks	AE	0.3414±0.0242	0.3415±0.0255	0.3416±0.0258
	GeoAE	0.0013±0.0003	0.0013±0.0003	0.0012±0.0002
	TopoAE	0.2159±0.0101	0.2143±0.0091	0.2140±0.0105
Bouncing rings	AE	0.2967±0.0485	0.2949±0.0456	0.2946±0.0447
	GeoAE	0.0013±0.0002	0.0014±0.0002	0.0013±0.0001
	TopoAE	0.2103±0.0120	0.2104±0.0104	0.2091±0.0102
Orbiting disks	AE	0.0052±0.0018	0.0051±0.0016	0.0050±0.0018
	GeoAE	0.0017±0.0001	0.0016±0.0001	0.0017±0.0001
	TopoAE	0.2280±0.0124	0.2209±0.0094	0.2227±0.0114
Orbiting rings	AE	0.0045±0.0014	0.0047±0.0016	0.0044±0.0014
	GeoAE	0.0017±0.0001	0.0019±0.0002	0.0017±0.0001
	TopoAE	0.2327±0.0151	0.2280±0.0142	0.2287±0.0128
Moving MNIST	AE	0.0048±0.0019	0.0047±0.0018	0.0044±0.0016
	GeoAE	0.0009±0.0001	0.0011±0.0001	0.0010±0.0001
	TopoAE	0.2868±0.0056	0.2877±0.0048	0.2823±0.0069
Lorenz-96	AE	0.1537±0.0064	0.1454±0.0089	0.1456±0.0061
	GeoAE	0.0011±0.0004	0.0016±0.0004	0.0014±0.0002
	TopoAE	0.2719±0.0298	0.2734±0.0371	0.2709±0.0322
ElectricDevices	AE	0.0136±0.0072	0.0139±0.0063	0.0111±0.0044
	GeoAE	0.0014±0.0001	0.0015±0.0002	0.0014±0.0002
	TopoAE	0.0610±0.0052	0.0631±0.0064	0.0616±0.0045
HeLa	AE	0.0437±0.0107	0.0409±0.0085	0.0399±0.0088
	GeoAE	0.0071±0.0035	0.0078±0.0038	0.0072±0.0039
	TopoAE	0.0589±0.0354	0.0559±0.0339	0.0559±0.0317
Glioblastoma	AE	0.0027±0.0010	0.0026±0.0005	0.0023±0.0006
	GeoAE	0.0011±0.0001	0.0013±0.0005	0.0010±0.0002
	TopoAE	0.0038±0.0013	0.0039±0.0010	0.0036±0.0012

4.3 Representation fidelity.

Table 5 summarizes input-to-latent metric and persistence distortion across encoders introduced in Section 3.3. GeoAE substantially reduces metric and H_0 persistence distortion relative to AE and TopoAE, indicating better preservation of pairwise geometry and connected-component structure. In contrast, H_1 distortion is low and similar across the three encoders. Metric distortion is strongly associated with H_0 distortion ($\rho = 0.837$, $p < 0.001$), but not with H_1 distortion ($\rho = 0.004$, $p = 0.962$). None of these fidelity measures significantly predicts the forecasting gain from explicit H_1 features. Thus, faithful input-topology preservation and predictively useful topology are distinct properties. We therefore treat fidelity as a representation diagnostic, rather than as the mechanism underlying forecasting gains. Full results are reported in Appendix A.2.

4.4 Effect of Geometry Regularization

Fig. 5 shows the empirical triangle behaviour: PCA rank with λ_{GeoAE} , MSE with PCA rank, and MSE with λ_{GeoAE} . It shows how the mean PCA effective rank of the latents varies the geometry-regularization λ , and how the forecasting MSE varies with this effectiveness rank. Higher effective rank is associated with lower forecasting-error rank $\rho = -0.479$, $p < 0.001$. We treat this association as descriptive because observations from the same dataset are related. Dataset-specific PCA spectra, effective-rank trends, and forecasting relationships are reported in Appendix D.1 Figures 8-11.

Table 4: Latent-window controls after z-only hyperparameter selection. PCA, KPCA, graph-Laplacian, diffusion-map, and dimension-matched random Fourier features replace the persistence descriptor with non-topological summaries computed from the same latent window. “Best topology” is the best persistence descriptor among the evaluated direct and fused topology variants. Lower MSE is better.

Dataset	Encoder	z	Fuse PCA	Fuse KPCA	Best RFF	Best spectral	Fuse H_1	Best topology
Bouncing disks	AE	0.3429±0.0266	0.4270±0.0273	0.3644±0.0277	0.3437±0.0282	0.3453±0.0291 (Diffusion)	0.3639±0.0297	0.3402±0.0256 (H_1)
	GeoAE	0.0013±0.0003	0.0022±0.0007	0.0025±0.0003	0.0014±0.0003	0.0014±0.0002 (Laplacian)	0.0025±0.0003	0.0010±0.0002 (Landscape H_0)
	TopoAE	0.2158±0.0099	0.2258±0.0100	0.2217±0.0096	0.2177±0.0091	0.2152±0.0079 (Laplacian)	0.2259±0.0075	0.2070±0.0083 (Landscape $H_0 + H_1$)
Bouncing rings	AE	0.2972±0.0474	0.3902±0.0620	0.3104±0.0557	0.2937±0.0467	0.2986±0.0509 (Laplacian)	0.3112±0.0561	0.2955±0.0495 (Landscape H_1)
	GeoAE	0.0013±0.0001	0.0022±0.0003	0.0023±0.0003	0.0013±0.0002	0.0013±0.0003 (Laplacian)	0.0023±0.0002	0.0011±0.0002 (Landscape H_0)
	TopoAE	0.2104±0.0120	0.2187±0.0153	0.2148±0.0132	0.2082±0.0121	0.2096±0.0091 (Laplacian)	0.2160±0.0105	0.2024±0.0103 (PersLay H_0)
Orbiting disks	AE	0.0052±0.0018	0.0064±0.0023	0.0069±0.0027	0.0051±0.0015	0.0051±0.0016 (Laplacian)	0.0070±0.0028	0.0047±0.0016 (PersLay H_0)
	GeoAE	0.0017±0.0001	0.0016±0.0002	0.0016±0.0001	0.0017±0.0001	0.0017±0.0001 (Fuse Laplacian)	0.0015±0.0001	0.0011±0.0001 (PersLay H_0)
	TopoAE	0.2281±0.0124	0.2766±0.0071	0.2800±0.0105	0.2239±0.0079	0.2235±0.0107 (Diffusion)	0.2761±0.0162	0.2083±0.0076 (Landscape H_1)
Orbiting rings	AE	0.0045±0.0014	0.0055±0.0019	0.0056±0.0021	0.0047±0.0014	0.0044±0.0015 (Laplacian)	0.0056±0.0019	0.0040±0.0013 (PersLay H_0)
	GeoAE	0.0017±0.0001	0.0014±0.0001	0.0016±0.0001	0.0016±0.0001	0.0016±0.0001 (Fuse Laplacian)	0.0017±0.0001	0.0010±0.0001 (PersLay H_0)
	TopoAE	0.2321±0.0144	0.2723±0.0302	0.2653±0.0232	0.2337±0.0106	0.2326±0.0160 (Laplacian)	0.2617±0.0202	0.2254±0.0080 (Landscape H_1)
Moving MNIST	AE	0.0048±0.0019	0.0081±0.0033	0.0086±0.0028	0.0048±0.0020	0.0046±0.0019 (Laplacian)	0.0087±0.0028	0.0046±0.0018 ($H_0 + H_1$)
	GeoAE	0.0009±0.0001	0.0015±0.0005	0.0021±0.0005	0.0011±0.0002	0.0009±0.0001 (Diffusion)	0.0027±0.0005	0.0009±0.0001 (H_0)
	TopoAE	0.2868±0.0056	0.3000±0.0098	0.3041±0.0104	0.2903±0.0056	0.2895±0.0096 (Laplacian)	0.3037±0.0119	0.2748±0.0061 (Landscape $H_0 + H_1$)
Lorenz-96	AE	0.1536±0.0057	0.2081±0.0274	0.2042±0.0229	0.1441±0.0078	0.1505±0.0031 (Diffusion)	0.1998±0.0223	0.1420±0.0119 (PI H_1)
	GeoAE	0.0012±0.0003	0.0022±0.0003	0.0017±0.0003	0.0015±0.0001	0.0012±0.0002 (Diffusion)	0.0017±0.0005	0.0012±0.0001 (PersLay $H_0 + H_1$)
	TopoAE	0.2726±0.0298	0.4258±0.0256	0.3299±0.0254	0.2749±0.0300	0.2726±0.0309 (Diffusion)	0.3264±0.0258	0.2745±0.0394 (H_1)
ElectricDevices	AE	0.0139±0.0072	0.0172±0.0106	0.0170±0.0095	0.0124±0.0065	0.0118±0.0051 (Laplacian)	0.0167±0.0094	0.0102±0.0031 (PI H_1)
	GeoAE	0.0014±0.0001	0.0014±0.0002	0.0017±0.0002	0.0014±0.0001	0.0016±0.0003 (Laplacian)	0.0017±0.0001	0.0011±0.0002 ($H_0 + H_1$)
	TopoAE	0.0615±0.0047	0.1205±0.0105	0.1070±0.0167	0.0626±0.0080	0.0604±0.0053 (Diffusion)	0.1210±0.0144	0.0492±0.0025 (Landscape $H_0 + H_1$)
HeLa	AE	0.0437±0.0107	0.0480±0.0076	0.0488±0.0098	0.0418±0.0088	0.0409±0.0051 (Diffusion)	0.0496±0.0081	0.0383±0.0083 (Landscape H_1)
	GeoAE	0.0072±0.0032	0.0130±0.0044	0.0127±0.0041	0.0080±0.0033	0.0079±0.0038 (Laplacian)	0.0146±0.0033	0.0071±0.0033 (Landscape H_1)
	TopoAE	0.0587±0.0365	0.0640±0.0354	0.0675±0.0392	0.0605±0.0361	0.0558±0.0326 (Laplacian)	0.0644±0.0351	0.0550±0.0324 (Landscape H_1)
Glioblastoma	AE	0.0027±0.0010	0.0079±0.0025	0.0086±0.0022	0.0022±0.0005	0.0025±0.0009 (Diffusion)	0.0100±0.0024	0.0021±0.0006 (PersLay H_1)
	GeoAE	0.0011±0.0001	0.0066±0.0029	0.0062±0.0019	0.0017±0.0005	0.0012±0.0005 (Laplacian)	0.0063±0.0035	0.0008±0.0001 (Landscape H_0)
	TopoAE	0.0040±0.0013	0.0092±0.0031	0.0081±0.0017	0.0048±0.0015	0.0037±0.0011 (Diffusion)	0.0078±0.0015	0.0033±0.0008 (PersLay H_1)

Table 5: Input-to-latent representation fidelity by encoder. Entries are medians over datasets and random seeds. Metric distortion measures changes in normalized pairwise distances; H_0 and H_1 distortion are bottleneck distances between input- and latent-space persistence diagrams. Lower is better.

Encoder	Metric distortion	H_0 distortion	H_1 distortion
AE	0.3561	0.4103	0.0209
GeoAE	0.1055	0.1139	0.0273
TopoAE	0.3416	0.3923	0.0229

5 Discussion

Latent representations do not necessarily preserve the visible topological structure of their original input; for example, a loop in pixel space may be compressed or distorted by the encoder, especially when the latter is trained only for reconstruction. This explains why H_1 is not always the strongest latent-space signal, as loop information may not survive the encoding. In contrast, H_0 captures connectivity and coarse clustering structure, which is often more stable in latent space. This motivates both the TopoAE regularizer, following [39], and the use of multiple topology descriptors, rather than assuming that pixel-space topology is preserved automatically.

GeoAE performs strongly because its distance-preserving regularizer encourages the latent space to retain input space geometry. This may provide the predictor with a more useful coordinate system and make topology computed over the latent trajectory more informative. This is consistent with the hypothesis that preserving latent geometry produces more faithful latent point clouds, making their persistent topology more informative for forecasting. TopoAE is weaker in our setting because preserving coarse H_0 persistence does not necessarily preserve the metric and temporal details needed for next-latent prediction.

Across most datasets, GeoAE also exhibits the largest relative reduction from its z -only baseline after topology is added. This suggests that geometry regularization and persistent topology are complementary: organizing the latent points through pairwise-distance preservation may produce sliding-window point clouds whose topological summaries are more informative for forecasting.

We compare scenarios against the baseline case of inputting only z into the predictor. When using the Betti sequence as the only input to the predictor (without z), it consistently produces the weakest predictor of future latent states. This is expected because Betti sequences summarize connectivity and loop structure but omit important information, like position and intensity, needed to reconstruct the future latent vector.

One important limitation is that Vietoris-Rips persistence on a point cloud is permutation-invariant: it depends on pairwise distances between points, not on the order in which the latent states occur. This still allows temporal prediction to be possible, because a descriptor is computed for each sliding window and the LSTM models how these

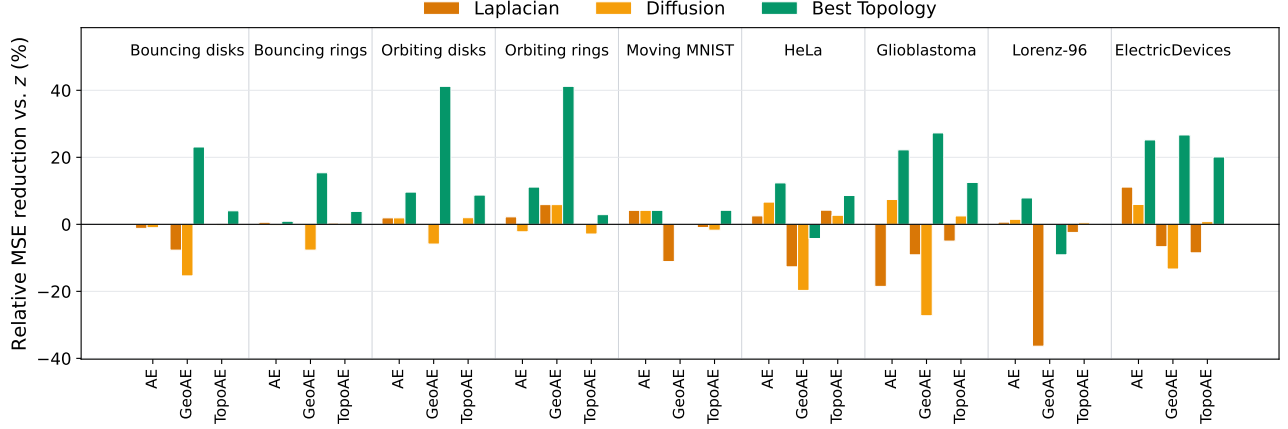


Figure 4: Relative MSE reduction compared with z -only forecasting for Laplacian, diffusion, and the best selected topology descriptor, across datasets and encoder variants

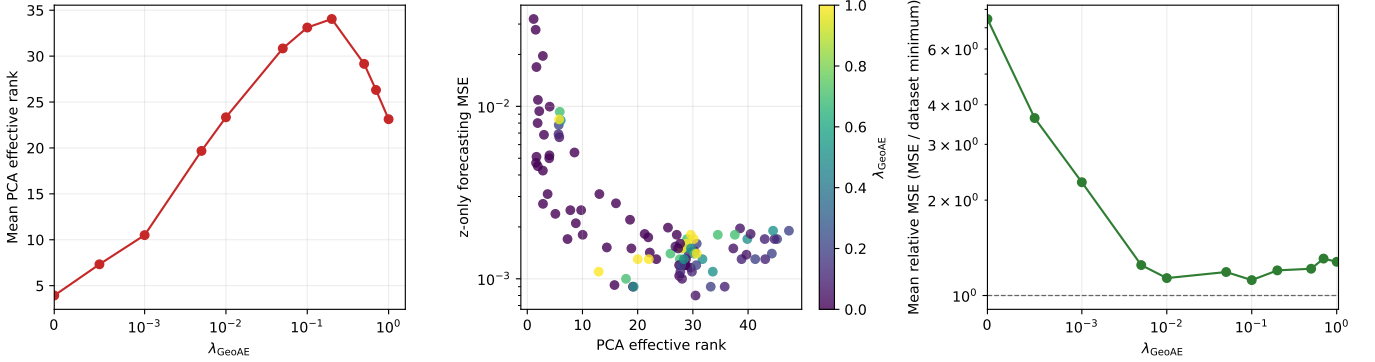


Figure 5: Effect of GeoAE regularization on latent dimensionality and forecasting. Left: mean PCA effective rank of the latent representation with respect to λ_{GeoAE} . Middle: raw z -only forecasting MSE versus PCA effective rank for each dataset- λ_{GeoAE} setting, coloured by regularization strength. Right: forecasting MSE at each λ_{GeoAE} , normalized by the lowest tested MSE within each dataset (to enable cross-dataset averaging). Values are averaged over 5 seeds.

window-level summaries evolve over time. The TDA descriptor contributes an order-invariant summary of the recent latent trajectory, while the recurrent predictor retains the temporal ordering between successive windows.

Non-topological controls test whether the gains are explained by generic summaries of the same latent window dimension. PCA and KPCA test linear and kernel variation, Laplacian eigenmaps and diffusion maps test manifold geometry, and dimension-matched RFFs test generic nonlinear expansion. These controls sometimes improve prediction, showing that latent spread and geometry are informative. However, topology remains stronger across most dataset-encoder settings, especially when landscapes or PersLay-style vectorizations are used, suggesting that the gains are not explained solely by adding dimensions or summarizing latent variation.

There is no universal ranking among the persistence vectorizations. Betti curves are coarse counts over filtration values, while persistence images, landscapes, and PersLay-style features retain different aspects of the birth-death distribution. Betti curves perform well in several individual settings, while landscapes and PersLay give the most consistent advantages against the stronger matched controls. This suggests that retaining more birth-death structure can help, though the best vectorizations are dataset- and encoder-dependent.

As an additional robustness check, Appendix Table 14 evaluates frozen V-JEPA2 features. This is a much larger general-purpose video encoder than our dataset-specific autoencoders, and is not optimized for our low-dimensional latent forecasting setting. Despite that, adding topology improves its forecasting performance, suggesting that the benefit of such auxiliary features is not specific to our encoder architecture.

Finally, the full sweep in Table 15 shows that topology-only inputs are consistently the weakest settings. Topological descriptors summarize structure but do not contain enough appearance, position, or intensity information to forecast

the next latent state on their own. The strongest results come from combining topology with z , supporting the view that topology provides complementary context, rather than a standalone representation.

We complement our experimental results with stability, recoverability, significance, and explainability analyses in Appendix A. The stability testing in Appendix A.5 shows that GeoAE gives the lowest H_0 ratio in 7/9 datasets and the lowest H_1 in 6/9 datasets. Its persistence diagrams therefore generally change by a smaller fraction of the latent point cloud displacement under input noise. This can occur because GeoAE regularizes pairwise distances, allowing latent coordinates to move while preserving enough relative geometry for VR persistence to remain comparatively stable. Because these quantities are normalized by d_H , they measure relative rather than absolute diagram stability. The recoverability analysis in Appendix A.6 shows that for most datasets, H_1 descriptors are not reliably predictable from z . Although the topology descriptor is deterministically computed from a latent window, it may make trajectory structure that is difficult for the chosen probe to recover directly, more accessible to the predictor. Glioblastoma is the exception, while ElectricDevices and HeLa show partial recoverability. Thus, topology may help by explicitly presenting structure that is difficult for a finite predictor to extract from the latent trajectory, or by making structure already present in z directly available to the predictor.

The significance analysis in Appendix A.3 compares fixed topology descriptors with fixed PCA, KPCA, Laplacian, diffusion, and RFF controls. All evaluated topology descriptors outperform PCA and KPCA in the paired comparisons. Against the stronger Laplacian, diffusion, and RFF controls, results depend on the vectorization: landscapes and PersLay provide the most consistent advantages. RFF is the strongest control: These one-sided p -values are uncorrected for multiple comparisons; the corresponding two-sided comparisons against RFF do not fall below 0.05. These results suggest that generic nonlinear expansion explains part, but not all, of the observed gains. The perturbation controls in Table 3 further show that corrupted topology descriptors can remain competitive, emphasizing that the gains should not be attributed solely to the exact persistence values.

Lastly, the explainability analysis in Appendix A.4 compares two-component PCA explainability with forecasting MSE. The percentage above each PCA panel in Figure 12 shows a positive within-dataset association between two-component PCA explainability and forecasting MSE. GeoAE generally has lower PCA explainability but lower MSE, suggesting that its predictive information is distributed across more latent dimensions, rather than concentrated in a visually compact two-dimensional representation.

5.1 Limitations

One limitation of incorporating topology is the cost of computing persistence features over sliding windows. Appendix Table 6 shows that persistence is slower than other controls, though it adds little accelerator memory overhead. GeoAE also requires additional pairwise distance regularization during training, but uses the same encoder architecture and parameter count as the standard AE at inference time.

Another limitation is that the gains depend on the encoder, dataset, persistence vectorization method, and window parameters, which adds components that require validation. Learned fusion does not consistently outperform direct augmentation in our experiments, meaning it may not always be beneficial.

Our conclusions are limited to 3 encoder families, 9 datasets, and next-latent MSE. VR persistence is also permutation-invariant within each window and so summarizes trajectory geometry without preserving its internal temporal order. Future work should evaluate larger encoder families, longer forecasting horizons or different tasks, decoded-frame quality, and more scalable or temporally aware topology constructions.

6 Conclusion

In this paper, we show that persistent homology descriptors provide complementary predictive structure for next latent forecasting. Across 9 datasets and 3 encoder families, such augmentations improve over the z baseline in 24/27 dataset-encoder settings, while topology-only inputs consistently perform worst. Thus, topology is most useful as auxiliary context rather than as replacement for the latent space. Geometry-preserving encoding consistently produces the strongest forecasting performance and the largest gains from topology augmentation, suggesting an empirical complementarity between geometry regularization and topological augmentation.

Matched PCA, KPCA, Laplacian, diffusion maps, and RFF controls show that the gains are not explained solely by additional dimensions manifold geometry, or generic nonlinear expansion. Stability and recoverability analyses suggest that the descriptors are relatively robust and are often not easily predicted from z alone. Significance tests show that the advantage depends on the vectorization: Betti curves remain competitive, while persistence landscapes and PersLay-style representations provide the most consistent improvements against the strongest controls.

Future work could explore many directions. One could consider more scalable or temporally aware persistence constructions, alternative filtrations such as sublevel-set persistence, and other fusion strategies. Different architectures

and predictors can be used, such as transformers, across additional modalities and tasks such as classification and anomaly detection. Finally, encoder regularization, latent dimension, topology-aware losses, distance functions, and decoded-frame quality are directions that would be useful to assess the best use-case of topological context.

References

- [1] Henry Adams et al. “Persistence images: A stable vector representation of persistent homology”. In: *Journal of Machine Learning Research* 18.8 (2017), pp. 1–35.
- [2] Mehmet E Aktas, Esra Akbas, and Ahmed El Fatmaoui. “Persistence homology of networks: methods and applications”. In: *Applied Network Science* 4.1 (2019), pp. 1–28.
- [3] Guillaume Alain and Yoshua Bengio. “Understanding intermediate layers using linear classifier probes”. In: *arXiv preprint arXiv:1610.01644* (2016).
- [4] Fatima Ali Aljanobi and Jeongkyu Lee. “Topological data analysis for classification of heart disease data”. In: *2021 IEEE International Conference on Big Data and Smart Computing (BigComp)*. IEEE, 2021, pp. 210–213.
- [5] Mahmoud Assran et al. “Self-Supervised Learning From Images With a Joint-Embedding Predictive Architecture”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. June 2023, pp. 15619–15629.
- [6] Mido Assran et al. “V-JEPA 2: Self-supervised video models enable understanding, prediction and planning”. In: *arXiv preprint arXiv:2506.09985* (2025).
- [7] Omri Azencot et al. “Forecasting sequential data using consistent Koopman autoencoders”. In: *International conference on machine learning*. PMLR, 2020, pp. 475–485.
- [8] Tadas Baltrušaitis, Chaitanya Ahuja, and Louis-Philippe Morency. “Multimodal machine learning: A survey and taxonomy”. In: *IEEE transactions on pattern analysis and machine intelligence* 41.2 (2018), pp. 423–443.
- [9] Danielle Barnes, Luis Polanco, and Jose A Perea. “A comparative study of machine learning methods for persistence diagrams”. In: *Frontiers in Artificial Intelligence* 4 (2021), p. 681174.
- [10] Ulrich Bauer. “Ripser: efficient computation of Vietoris-Rips persistence barcodes”. In: *J. Appl. Comput. Topol.* 5.3 (2021), pp. 391–423. ISSN: 2367-1726. DOI: 10.1007/s41468-021-00071-5. URL: <https://doi.org/10.1007/s41468-021-00071-5>.
- [11] Mikhail Belkin and Partha Niyogi. “Laplacian eigenmaps for dimensionality reduction and data representation”. In: *Neural computation* 15.6 (2003), pp. 1373–1396.
- [12] Dmitri Burago, Yuri Burago, Sergei Ivanov, et al. *A course in metric geometry*. Vol. 33. American Mathematical Society Providence, 2001.
- [13] Gunnar Carlsson. “Topology and data”. In: *Bulletin of the American Mathematical Society* 46.2 (2009), pp. 255–308.
- [14] Gunnar Carlsson et al. “Persistence barcodes for shapes”. In: *Proceedings of the 2004 Eurographics/ACM SIG-GRAPH symposium on Geometry processing*. 2004, pp. 124–135.
- [15] Mathieu Carrière et al. “Perslay: A neural network layer for persistence diagrams and new graph topological signatures”. In: *International Conference on Artificial Intelligence and Statistics*. PMLR, 2020, pp. 2786–2796.
- [16] Frédéric Chazal et al. *The Structure and Stability of Persistence Modules*. Springer, 2016.
- [17] Nutan Chen, Patrick van der Smagt, and Botond Cseke. “Local Distance Preserving Auto-encoders using Continuous kNN Graphs”. In: *Proceedings of Topological, Algebraic, and Geometric Learning Workshops 2022*. Ed. by Alexander Cloninger et al. Vol. 196. Proceedings of Machine Learning Research. PMLR, July 2022, pp. 55–66. URL: <https://proceedings.mlr.press/v196/chen22b.html>.
- [18] James R Clough et al. “A topological loss function for deep-learning based image segmentation using persistent homology”. In: *IEEE transactions on pattern analysis and machine intelligence* 44.12 (2020), pp. 8766–8778.
- [19] David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. “Stability of persistence diagrams”. In: *Proceedings of the twenty-first annual symposium on Computational geometry*. 2005, pp. 263–271.
- [20] Ronald R Coifman and Stéphane Lafon. “Diffusion maps”. In: *Applied and computational harmonic analysis* 21.1 (2006), pp. 5–30.
- [21] Hoang Anh Dau et al. *The UCR Time Series Classification Archive*. https://www.cs.ucr.edu/~eamonn/time_series_data_2018/. Oct. 2018.

- [22] Thomas Davies. “Topological data analysis for anomaly detection in host-based logs”. In: *arXiv preprint arXiv:2204.12919* (2022).
- [23] Herbert Edelsbrunner and John Harer. *Computational topology: an introduction*. American Mathematical Soc., 2010.
- [24] Gene H Golub and Charles F Van Loan. *Matrix computations*. JHU press, 2013.
- [25] Albert Gu et al. “Learning mixed-curvature representations in product spaces”. In: *International conference on learning representations*. 2018.
- [26] Jean-Claude Hausmann. “On the Vietoris–Rips Complexes and a Cohomology Theory”. In: *Prospects in topology: proceedings of a conference in honor of William Browder*. 138. Princeton University Press. 1995, p. 175.
- [27] Geoffrey E Hinton and Ruslan R Salakhutdinov. “Reducing the dimensionality of data with neural networks”. In: *science* 313.5786 (2006), pp. 504–507.
- [28] Sepp Hochreiter and Jürgen Schmidhuber. “Long short-term memory”. In: *Neural computation* 9.8 (1997), pp. 1735–1780.
- [29] Shizuo Kaji, Takeki Sudo, and Kazushi Ahara. “Cubical ripser: Software for computing persistent homology of image and volume data”. In: *arXiv preprint arXiv:2005.12692* (2020).
- [30] Alperen Karan and Atabey Kaygun. “Time series classification via topological data analysis”. In: *Expert Systems with Applications* 183 (2021), p. 115326. ISSN: 0957-4174. DOI: <https://doi.org/10.1016/j.eswa.2021.115326>. URL: <https://www.sciencedirect.com/science/article/pii/S0957417421007557>.
- [31] Namwoo Kim, Hyungryul Baik, and Yoonjin Yoon. “TopoCL: Topological contrastive learning for time series”. In: *IEEE Transactions on Neural Networks and Learning Systems* (2026).
- [32] Jennifer Kloke and Gunnar Carlsson. “Topological de-noising: Strengthening the topological signal”. In: *arXiv preprint arXiv:0910.5947* (2009).
- [33] Aditi S Krishnapriyan, Maciej Haranczyk, and Dmitriy Morozov. “Topological descriptors help predict guest adsorption in nanoporous materials”. In: *The Journal of Physical Chemistry C* 124.17 (2020), pp. 9360–9368.
- [34] Yann LeCun et al. “Gradient-based learning applied to document recognition”. In: *Proceedings of the IEEE* 86.11 (1998), pp. 2278–2324.
- [35] Bowen Liu et al. “Deep Learning in Latent Space for Video Prediction and Compression”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. June 2021, pp. 701–710.
- [36] Edward N Lorenz. “Predictability: A problem partly solved”. In: *Proc. Seminar on predictability*. Vol. 1. 1. Reading. 1996, pp. 1–18.
- [37] Sourav Majumdar and Arnab Kumar Laha. “Clustering and classification of time series using topological data analysis with applications to finance”. In: *Expert Systems with Applications* 162 (2020), p. 113868.
- [38] Martin Maška et al. “The cell tracking challenge: 10 years of objective benchmarking”. In: *Nature Methods* 20.7 (2023), pp. 1010–1020.
- [39] Michael Moor et al. “Topological Autoencoders”. In: *Proceedings of the 37th International Conference on Machine Learning (ICML)*. Ed. by Hal Daumé III and Aarti Singh. Vol. 119. Proceedings of Machine Learning Research. PMLR, 2020, pp. 7045–7054. arXiv: 1906.00722 [cs.LG]. URL: <http://proceedings.mlr.press/v119/moor20a.html>.
- [40] Nina Otter et al. “A roadmap for the computation of persistent homology”. In: *EPJ data science* 6.1 (2017), p. 17.
- [41] Pratyush Pranav et al. “The topology of the cosmic web in terms of persistent Betti numbers”. In: *Monthly Notices of the Royal Astronomical Society* 465.4 (2017), pp. 4281–4310.
- [42] Ali Rahimi and Benjamin Recht. “Random features for large-scale kernel machines”. In: *Advances in neural information processing systems* 20 (2007).
- [43] Nathaniel Saul and Chris Tralie. *Scikit-TDA: Topological Data Analysis for Python*. 2019. DOI: 10.5281/zenodo.2533369.
- [44] Bernhard Schölkopf, Alexander Smola, and Klaus-Robert Müller. “Nonlinear component analysis as a kernel eigenvalue problem”. In: *Neural computation* 10.5 (1998), pp. 1299–1319.
- [45] Charles Spearman. *The proof and measurement of association between two things*. Appleton-Century-Crofts, 1961.

- [46] Nitish Srivastava, Elman Mansimov, and Ruslan Salakhudinov. “Unsupervised learning of video representations using lstms”. In: *International conference on machine learning*. PMLR, 2015, pp. 843–852.
- [47] Floris Takens. “Detecting strange attractors in turbulence”. In: *Dynamical Systems and Turbulence, Warwick 1980* (1981), pp. 366–381.
- [48] The GUDHI Project. *GUDHI User and Reference Manual*. GUDHI Editorial Board, 2015. URL: <http://gudhi.gforge.inria.fr/doc/latest/>.
- [49] Michael E Tipping and Christopher M Bishop. “Probabilistic principal component analysis”. In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 61.3 (1999), pp. 611–622.
- [50] Christopher J. Tralie and Jose A. Perea. “(Quasi)Periodicity Quantification in Video Data, Using Topology”. In: *SIAM Journal on Imaging Sciences* 11.2 (2018), pp. 1049–1077. DOI: 10.1137/17M1150736. eprint: <https://doi.org/10.1137/17M1150736>. URL: <https://doi.org/10.1137/17M1150736>.
- [51] Sarah Tymochko et al. “Using persistent homology to quantify a diurnal cycle in hurricanes”. In: *Pattern Recognition Letters* 133 (2020), pp. 137–143.
- [52] Yuhei Umeda. “Time series classification via topological data analysis”. In: *Information and Media Technologies* 12 (2017), pp. 228–239.
- [53] Ulrike Von Luxburg. “A tutorial on spectral clustering”. In: *Statistics and computing* 17.4 (2007), pp. 395–416.
- [54] David Williams. *Probability with martingales*. Cambridge university press, 1991.
- [55] Robert F Woolson. “Wilcoxon signed-rank test”. In: *Wiley encyclopedia of clinical trials* (2007), pp. 1–3.
- [56] Xiaojun Zheng et al. “Percept: a new online change-point detection method using topological data analysis”. In: *Technometrics* 65.2 (2023), pp. 162–178.
- [57] Afra Zomorodian and Gunnar Carlsson. “Computing persistent homology”. In: *Discrete & Computational Geometry* 33.2 (2005), pp. 249–274.

A Additional Tests

This section contains additional tests to support the experiments, mainly the profiling, the stability diagnosis, the recoverability, and the significance test.

A.1 Profiling

Table 6: GPU profiling summary at profile size 128, averaged over datasets after discarding profiling warmup runs. Runtime, added time, peak CPU RSS, peak accelerator memory, and phase times are reported as mean $\pm$ std across datasets and measured repeats. Added time and Runtime/ z are relative to the corresponding z -only run within the same dataset, encoder family, and profiling repeat. Persistence is CPU-side in our implementation.

Encoder	Input mode	Runtime (s)	Added time (s)	Runtime/ z	Peak memory (MB)		Phase time (s)			
					CPU RSS	Accel.	Enc.	Feature/fuse	Persist.	Pred.
AE	z	0.05 $\pm$ 0.02	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	4047 $\pm$ 2020	134 $\pm$ 19	0.03 $\pm$ 0.02	–	–	0.00 $\pm$ 0.00
AE	z + PCA	0.36 $\pm$ 0.24	0.31 $\pm$ 0.22	7.00 $\pm$ 4.28	4051 $\pm$ 2014	134 $\pm$ 19	0.03 $\pm$ 0.02	0.30 $\pm$ 0.22	–	0.00 $\pm$ 0.00
AE	z + KPCA	0.51 $\pm$ 0.35	0.46 $\pm$ 0.33	10.05 $\pm$ 6.31	4051 $\pm$ 2014	134 $\pm$ 19	0.03 $\pm$ 0.02	0.46 $\pm$ 0.33	–	0.00 $\pm$ 0.00
AE	z_{fuse} + H_1	1.74 $\pm$ 1.19	1.69 $\pm$ 1.17	33.81 $\pm$ 23.22	4057 $\pm$ 2008	138 $\pm$ 18	0.03 $\pm$ 0.02	0.00 $\pm$ 0.00	1.31 $\pm$ 0.90	0.01 $\pm$ 0.00
AE	z_{fuse} + PI	2.02 $\pm$ 1.37	1.97 $\pm$ 1.35	39.40 $\pm$ 26.88	4058 $\pm$ 2007	139 $\pm$ 19	0.03 $\pm$ 0.02	0.28 $\pm$ 0.18	1.31 $\pm$ 0.90	0.01 $\pm$ 0.00
AE	z_{fuse} + Landscape	1.86 $\pm$ 1.27	1.82 $\pm$ 1.25	36.29 $\pm$ 24.56	4058 $\pm$ 2007	140 $\pm$ 19	0.03 $\pm$ 0.02	0.13 $\pm$ 0.08	1.30 $\pm$ 0.90	0.01 $\pm$ 0.00
AE	z_{fuse} + PersLay	1.82 $\pm$ 1.21	1.77 $\pm$ 1.20	35.48 $\pm$ 23.80	4058 $\pm$ 2007	140 $\pm$ 19	0.03 $\pm$ 0.02	0.08 $\pm$ 0.06	1.31 $\pm$ 0.89	0.01 $\pm$ 0.00
AE	z_{fuse} + PCA	0.35 $\pm$ 0.24	0.31 $\pm$ 0.22	6.98 $\pm$ 4.30	4058 $\pm$ 2007	138 $\pm$ 18	0.03 $\pm$ 0.02	0.30 $\pm$ 0.22	–	0.01 $\pm$ 0.00
AE	z_{fuse} + KPCA	0.51 $\pm$ 0.35	0.47 $\pm$ 0.33	10.05 $\pm$ 6.31	4061 $\pm$ 2003	138 $\pm$ 18	0.03 $\pm$ 0.02	0.46 $\pm$ 0.33	–	0.01 $\pm$ 0.00
AE	z_{fuse} + Laplacian	0.72 $\pm$ 0.49	0.67 $\pm$ 0.47	14.03 $\pm$ 8.64	4061 $\pm$ 2003	138 $\pm$ 18	0.03 $\pm$ 0.02	0.66 $\pm$ 0.47	–	0.01 $\pm$ 0.00
AE	z_{fuse} + Diffusion	0.69 $\pm$ 0.47	0.64 $\pm$ 0.45	13.33 $\pm$ 8.38	4061 $\pm$ 2003	138 $\pm$ 18	0.03 $\pm$ 0.02	0.63 $\pm$ 0.45	–	0.01 $\pm$ 0.00
AE	z_{fuse} + RFF	0.25 $\pm$ 0.16	0.20 $\pm$ 0.14	4.93 $\pm$ 2.72	4061 $\pm$ 2003	138 $\pm$ 18	0.03 $\pm$ 0.02	0.20 $\pm$ 0.14	–	0.01 $\pm$ 0.00
GeoAE	z	0.05 $\pm$ 0.02	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	4061 $\pm$ 2003	134 $\pm$ 19	0.03 $\pm$ 0.02	–	–	0.00 $\pm$ 0.00
GeoAE	z + PCA	0.36 $\pm$ 0.24	0.31 $\pm$ 0.22	6.95 $\pm$ 4.40	4061 $\pm$ 2003	134 $\pm$ 19	0.03 $\pm$ 0.02	0.31 $\pm$ 0.22	–	0.00 $\pm$ 0.00
GeoAE	z + KPCA	0.52 $\pm$ 0.36	0.47 $\pm$ 0.34	9.95 $\pm$ 6.53	4061 $\pm$ 2003	134 $\pm$ 19	0.03 $\pm$ 0.02	0.47 $\pm$ 0.34	–	0.00 $\pm$ 0.00
GeoAE	z_{fuse} + H_1	1.81 $\pm$ 1.24	1.76 $\pm$ 1.22	34.80 $\pm$ 25.45	4061 $\pm$ 2003	138 $\pm$ 18	0.03 $\pm$ 0.02	0.00 $\pm$ 0.00	1.37 $\pm$ 0.95	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + PI	2.11 $\pm$ 1.42	2.06 $\pm$ 1.40	40.80 $\pm$ 29.51	4061 $\pm$ 2003	139 $\pm$ 19	0.03 $\pm$ 0.02	0.30 $\pm$ 0.19	1.37 $\pm$ 0.94	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + Landscape	1.95 $\pm$ 1.32	1.91 $\pm$ 1.30	37.77 $\pm$ 27.16	4061 $\pm$ 2003	140 $\pm$ 19	0.03 $\pm$ 0.02	0.14 $\pm$ 0.09	1.37 $\pm$ 0.94	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + PersLay	1.96 $\pm$ 1.29	1.91 $\pm$ 1.27	38.01 $\pm$ 26.94	4061 $\pm$ 2003	139 $\pm$ 19	0.03 $\pm$ 0.02	0.14 $\pm$ 0.10	1.40 $\pm$ 0.95	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + PCA	0.36 $\pm$ 0.24	0.31 $\pm$ 0.22	7.01 $\pm$ 4.41	4061 $\pm$ 2003	138 $\pm$ 18	0.03 $\pm$ 0.02	0.31 $\pm$ 0.22	–	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + KPCA	0.52 $\pm$ 0.35	0.47 $\pm$ 0.33	9.97 $\pm$ 6.47	4061 $\pm$ 2003	138 $\pm$ 18	0.03 $\pm$ 0.02	0.47 $\pm$ 0.33	–	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + Laplacian	0.72 $\pm$ 0.48	0.67 $\pm$ 0.46	13.70 $\pm$ 8.79	4064 $\pm$ 2000	138 $\pm$ 18	0.03 $\pm$ 0.02	0.66 $\pm$ 0.46	–	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + Diffusion	0.68 $\pm$ 0.46	0.63 $\pm$ 0.44	13.06 $\pm$ 8.15	4064 $\pm$ 2000	138 $\pm$ 18	0.03 $\pm$ 0.02	0.63 $\pm$ 0.44	–	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + RFF	0.25 $\pm$ 0.15	0.20 $\pm$ 0.14	4.85 $\pm$ 2.68	4064 $\pm$ 2000	138 $\pm$ 18	0.03 $\pm$ 0.02	0.19 $\pm$ 0.14	–	0.01 $\pm$ 0.00
TopoAE	z	0.05 $\pm$ 0.02	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	4064 $\pm$ 2000	134 $\pm$ 19	0.03 $\pm$ 0.02	–	–	0.00 $\pm$ 0.00
TopoAE	z + PCA	0.35 $\pm$ 0.23	0.30 $\pm$ 0.22	6.81 $\pm$ 4.29	4064 $\pm$ 2000	134 $\pm$ 19	0.03 $\pm$ 0.02	0.30 $\pm$ 0.22	–	0.00 $\pm$ 0.00
TopoAE	z + KPCA	0.51 $\pm$ 0.35	0.46 $\pm$ 0.33	9.88 $\pm$ 6.67	4064 $\pm$ 2000	134 $\pm$ 19	0.03 $\pm$ 0.02	0.46 $\pm$ 0.33	–	0.00 $\pm$ 0.00
TopoAE	z_{fuse} + H_1	1.75 $\pm$ 1.21	1.70 $\pm$ 1.20	33.50 $\pm$ 24.59	4064 $\pm$ 2000	139 $\pm$ 19	0.03 $\pm$ 0.02	0.00 $\pm$ 0.00	1.31 $\pm$ 0.92	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + PI	2.00 $\pm$ 1.38	1.96 $\pm$ 1.36	38.51 $\pm$ 27.87	4064 $\pm$ 1999	139 $\pm$ 19	0.03 $\pm$ 0.02	0.28 $\pm$ 0.18	1.30 $\pm$ 0.91	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + Landscape	1.87 $\pm$ 1.29	1.82 $\pm$ 1.27	35.84 $\pm$ 26.15	4064 $\pm$ 1999	141 $\pm$ 20	0.03 $\pm$ 0.02	0.13 $\pm$ 0.08	1.30 $\pm$ 0.90	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + PersLay	1.82 $\pm$ 1.24	1.77 $\pm$ 1.22	35.05 $\pm$ 25.28	4065 $\pm$ 1999	139 $\pm$ 19	0.03 $\pm$ 0.02	0.08 $\pm$ 0.06	1.31 $\pm$ 0.91	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + PCA	0.36 $\pm$ 0.24	0.31 $\pm$ 0.22	6.86 $\pm$ 4.27	4068 $\pm$ 1995	138 $\pm$ 18	0.03 $\pm$ 0.02	0.30 $\pm$ 0.22	–	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + KPCA	0.52 $\pm$ 0.35	0.47 $\pm$ 0.33	9.93 $\pm$ 6.66	4071 $\pm$ 1992	138 $\pm$ 18	0.03 $\pm$ 0.02	0.46 $\pm$ 0.33	–	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + Laplacian	0.72 $\pm$ 0.49	0.67 $\pm$ 0.47	13.69 $\pm$ 9.01	4071 $\pm$ 1992	138 $\pm$ 18	0.03 $\pm$ 0.02	0.66 $\pm$ 0.47	–	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + Diffusion	0.69 $\pm$ 0.46	0.64 $\pm$ 0.45	13.13 $\pm$ 8.51	4072 $\pm$ 1991	138 $\pm$ 18	0.03 $\pm$ 0.02	0.64 $\pm$ 0.45	–	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + RFF	0.25 $\pm$ 0.15	0.20 $\pm$ 0.13	4.79 $\pm$ 2.67	4079 $\pm$ 1983	138 $\pm$ 18	0.03 $\pm$ 0.02	0.19 $\pm$ 0.13	–	0.01 $\pm$ 0.00

We profile inference-time cost using previously-trained encoders and predictors, discarding one warmup run and averaging over 3 measured repeats. Profile size 128 means each run is capped at 128 training sequences and 128 test sequences. Table 6 shows that topology overhead is dominated by the persistence phase: PCA/KPCA add approximately 0.3-0.5 seconds, RFF adds 0.2 seconds, and Laplacian and diffusion map adds 0.6-0.7 seconds, whereas persistence-based inputs add 1.7-2.1 seconds per run. Accelerator-memory overhead is small and runtime variance is mostly due to dataset size and sequence length.

A.2 Representation Fidelity and Distortion

To evaluate input-to-latent representation fidelity, we sampled 100 fixed test trajectory windows for every dataset, encoder, and random seed. For each window, we computed pairwise Euclidean distance matrices between the vectorized input frames and their latent representations, normalizing each matrix by its mean off-diagonal distance. Metric distortion was measured as the root-mean-square difference between corresponding normalized distances. We then computed Vietoris–Rips persistence diagrams directly from both distance matrices and measured H_0 and H_1 persistence distortion using bottleneck distance. Window-level values were aggregated by their median, yielding one fidelity

Table 7: Input-to-latent representation fidelity for each dataset and encoder. Entries are medians over five random seeds. Metric distortion measures changes in normalized pairwise distances; H_0 and H_1 distortion are bottleneck distances between input- and latent-space persistence diagrams. Lower is better.

Dataset	Encoder	Metric distortion	H_0 distortion	H_1 distortion
Bouncing rings	AE	0.3362	0.4399	0.0209
	GeoAE	0.1053	0.0947	0.0273
	TopoAE	0.3390	0.4399	0.0229
Bouncing disks	AE	0.2433	0.3647	0.0327
	GeoAE	0.1005	0.1243	0.0346
	TopoAE	0.2947	0.3774	0.0301
Orbiting rings	AE	0.4422	0.4460	0.0324
	GeoAE	0.1055	0.1297	0.0280
	TopoAE	0.3420	0.4462	0.0375
Orbiting disks	AE	0.3418	0.4103	0.0513
	GeoAE	0.0998	0.1013	0.0404
	TopoAE	0.3416	0.4192	0.0524
Moving MNIST	AE	0.3561	0.4401	0.0313
	GeoAE	0.1176	0.1392	0.0329
	TopoAE	0.3719	0.4554	0.0298
Lorenz-96	AE	0.0947	0.0471	0.0000
	GeoAE	0.0465	0.0358	0.0000
	TopoAE	0.0971	0.0459	0.0000
ElectricDevices	AE	0.4102	0.3906	0.0000
	GeoAE	0.1076	0.1034	0.0104
	TopoAE	0.3731	0.3821	0.0000
Glioblastoma	AE	0.6345	0.3699	0.0071
	GeoAE	0.1183	0.1139	0.0192
	TopoAE	0.6565	0.3923	0.0065
HeLa	AE	0.5363	0.4555	0.0101
	GeoAE	0.1528	0.1809	0.0191
	TopoAE	0.2707	0.2939	0.0136

measurement for each dataset–encoder–seed combination.

Table 7 reports metric, H_0 , and H_1 distortion for every dataset and encoder. GeoAE consistently exhibits lower metric and H_0 distortion, whereas H_1 distortion is generally low for all encoders.

We compared these measurements with the relative forecasting gain from augmenting z with H_1 features using Spearman rank correlation.

Table 8 shows a strong association between metric and H_0 distortion, but no significant association involving H_1 distortion or forecasting gain. As a robustness check against dependence among random seeds, we repeated the analysis after aggregating seeds within each dataset–encoder combination ($n = 27$); the conclusions were unchanged.

A.3 Significance Testing

To test whether topology improves over non-topological geometric controls, we compare fixed topology descriptors against fixed PCA, KPCA, Laplacian, diffusion-map, and RFF descriptors using paired MSE values. For each descriptor, we first average MSE over encoder families within each dataset, producing one paired comparison per dataset, then apply a one-sided Wilcoxon signed-rank test over the 9 datasets [55]. This avoids treating encoder variants as independent samples and avoids selecting the best method based on test performance.

Table 9, in conjunction with Fig. 4, shows that the topology advantage depends on both the persistence vectorization and the control. All topology descriptors strongly outperform PCA and KPCA, while persistence landscapes and PersLay also outperform the Laplacian and diffusion controls. RFF is the strongest control: Betti curves and persistence images are not significantly better than RFF, while landscapes and PersLay retain smaller one-sided advantages ($p = 0.0273$) and ($p = 0.0371$). Thus, generic nonlinear expansion explains part of the gain, but richer persistence vectorizations provide the most consistent advantage across the matched controls.

Table 8: Spearman associations between representation distortion, persistence fidelity, and the relative forecasting gain from augmenting z with H_1 . Correlations use dataset–encoder–seed observations ($n = 135$).

Association	ρ	p
Metric distortion vs. H_0 distortion	0.837	< 0.001
Metric distortion vs. H_1 distortion	0.004	0.962
Metric distortion vs. forecasting gain	0.064	0.462
H_0 distortion vs. forecasting gain	0.040	0.646
H_1 distortion vs. forecasting gain	0.064	0.462

Table 9: Fixed paired comparisons of persistence descriptors with non-topological controls. Wins are reported over all 27 dataset–encoder settings, whereas statistical tests use nine independent dataset averages to avoid treating encoder variants as independent samples. Lower MSE is better.

Topology descriptor	Wins	Avg. gain	Wilcoxon p	Two-sided p
Betti H_1 vs. PCA	25/27	25.9%	0.0020	0.0039
Betti H_1 vs. KPCA	26/27	21.5%	0.0020	0.0039
Betti H_1 vs. Laplacian	14/27	2.1%	0.0039	0.0078
Betti H_1 vs. Diffusion	14/27	0.6%	0.0195	0.0391
Betti H_1 vs. RFF	14/27	1.7%	0.1016	0.2031
PI H_1 vs. PCA	23/27	24.3%	0.0020	0.0039
PI H_1 vs. KPCA	24/27	20.0%	0.0020	0.0039
PI H_1 vs. Laplacian	13/27	-1.3%	0.1250	0.2500
PI H_1 vs. Diffusion	12/27	-3.1%	0.1250	0.2500
PI H_1 vs. RFF	12/27	-1.5%	0.4102	0.8203
Landscape H_1 vs. PCA	25/27	28.7%	0.0020	0.0039
Landscape H_1 vs. KPCA	26/27	24.5%	0.0020	0.0039
Landscape H_1 vs. Laplacian	18/27	6.4%	0.0020	0.0039
Landscape H_1 vs. Diffusion	16/27	5.1%	0.0059	0.0117
Landscape H_1 vs. RFF	14/27	6.1%	0.0273	0.0547
PersLay H_1 vs. PCA	24/27	27.5%	0.0020	0.0039
PersLay H_1 vs. KPCA	25/27	23.2%	0.0020	0.0039
PersLay H_1 vs. Laplacian	16/27	5.2%	0.0020	0.0039
PersLay H_1 vs. Diffusion	18/27	3.8%	0.0020	0.0039
PersLay H_1 vs. RFF	15/27	4.8%	0.0371	0.0742

A.4 Explainability

Each encoder transforms the same observations X into a distinct latent geometry. PCA explainability is the percentage of total latent variance captured by PC1 and PC2; a higher value indicates the latent representation is concentrated along fewer dominant directions. Note that this does not mean the dataset itself is simpler. Figure 12 shows the PCA components for each dataset and encoder. GeoAE generally has a lower PCA explainability but lower forecasting MSE, whereas TopoAE is often more concentrated in two dimensions yet predicts worse.

As summarized in Table 10, PCA explainability is positively associated with forecasting MSE in every dataset (Spearman $\rho = 0.5$ or 1.0) [45]. So, concentration into a few high-variance directions does not imply better predictability; GeoAE may instead distribute predictive information across more latent dimensions, though this association does not imply causality.

A.5 Stability Diagnosis

Our topological descriptors are computed from sliding-window point clouds of latent vectors using VR persistence. Persistence diagrams are stable summaries of topology under suitable perturbations [19]; in our setting, where diagrams are computed from latent point clouds using VR filtrations, the corresponding point-cloud stability result relates Hausdorff perturbations of the latent cloud to bottleneck perturbations of the diagram (ie, if two point clouds are close in Hausdorff distance, then their persistence diagrams are close in bottleneck distance) [16]. For a latent point cloud P , let $\text{Dgm}_k(P)$ denote the k -dimensional persistence diagram obtained from its VR filtration. This does not imply stability of the full forecasting pipeline, but implies that the persistence step itself does not arbitrarily amplify small

Table 10: Two-component PCA explainability and z -forecasting MSE across encoders. Each PCA value is the percentage of latent variance explained by PC1 and PC2. The final column reports Spearman’s ρ

Dataset	AE		GeoAE		TopoAE		Spearman ρ
	PC1+PC2 (%)	MSE	PC1+PC2 (%)	MSE	PC1+PC2 (%)	MSE	
Bouncing disks	52.0	0.3429	21.3	0.0013	47.5	0.2158	1.00
Bouncing rings	75.8	0.2972	18.3	0.0013	49.9	0.2104	1.00
Orbiting disks	93.6	0.0052	13.7	0.0017	42.7	0.2281	0.50
Orbiting rings	94.3	0.0045	15.4	0.0017	44.6	0.2321	0.50
Moving MNIST	89.8	0.0048	35.8	0.0009	48.3	0.2868	0.50
Lorenz-96	33.2	0.1536	14.6	0.0012	25.3	0.2726	0.50
ElectricDevices	99.6	0.0139	29.4	0.0014	64.7	0.0615	0.50
Glioblastoma	49.8	0.0027	28.1	0.0011	59.9	0.0040	1.00
HeLa	97.9	0.0437	72.9	0.0072	86.8	0.0587	0.50

perturbations of the latent point cloud. For finite latent point clouds $P, Q \subset \mathbb{R}^d$, we use the Hausdorff distance d_H [12], which measures how far two point clouds are from each other

$$d_H(P, Q) = \max \left\{ \max_{p \in P} \min_{q \in Q} \|p - q\|_2, \max_{q \in Q} \min_{p \in P} \|q - p\|_2 \right\},$$

where $\|\cdot\|_2$ is Euclidean distance in latent space. The bottleneck distance d_B between two persistence diagrams $\text{Dgm}_k(P)$ and $\text{Dgm}_k(Q)$ is [23]

$$d_B(\text{Dgm}_k(P), \text{Dgm}_k(Q)) = \inf_{\gamma: \text{Dgm}_k(P) \rightarrow \text{Dgm}_k(Q)} \sup_{x \in \text{Dgm}_k(P)} \|x - \gamma(x)\|_\infty,$$

where γ ranges over matchings of birth–death points, allowing unmatched points to be matched to the diagonal. For Vietoris–Rips persistence, point-cloud stability gives

$$d_B(\text{Dgm}_k(P), \text{Dgm}_k(Q)) \leq C d_H(P, Q),$$

for homology dimension k , up to a filtration-dependent constant C [16].

To test this stability empirically, we perturb the input frames X with additive noise and compare the clean and noisy latent point clouds. Let $P_t = \{z_{t-w+1}, \dots, z_t\}$ be the latent window from clean frames and P'_t the corresponding window after noising the input frames and re-encoding them. We measure the latent perturbation by the Hausdorff distance $d_H(P_t, P'_t)$, and the topological perturbation by the bottleneck distance $d_B(\text{Dgm}_k(P_t), \text{Dgm}_k(P'_t))$. We report the ratio

$$\rho_k(t) = \frac{d_B(\text{Dgm}_k(P_t), \text{Dgm}_k(P'_t))}{d_H(P_t, P'_t)}.$$

Values $\rho_k(t) < 1$ indicate that the persistence diagram changes less than the latent point cloud does, under the same perturbation.

Appendix Table 11 evaluates the stability of the latent persistence branch under input noise σ . When $\sigma = 0.10$, all reported ratios are below 1, meaning persistence diagrams change less than the latent point clouds themselves. In particular, the H_1 ratios $\rho_1 = d_B(\text{Dgm}_1(P_t), \text{Dgm}_1(P'_t))/d_H(P_t, P'_t)$ are consistently smaller than the H_0 ratios, suggesting that loop-level structure is less sensitive to input noise than connected-component structure in these latent point clouds.

GeoAE often exhibits larger latent point cloud displacement d_H under input noise, but yields the lowest relative H_0 and H_1 changes in 7/9 and 6/9 datasets respectively. So, its persistence diagrams generally change by a smaller fraction of the corresponding latent displacement. This may occur because GeoAE regularizes pairwise distances in latent space: noise can move the latent coordinates, increasing (d_H) , while preserving much of the relative geometry of the point cloud. Since VR persistence depends on pairwise distances rather than absolute coordinates, the topology can remain relatively stable even when the latent vectors move. These ratios measure stability relative to (d_H) , rather than absolute diagram change.

A.6 Recoverability Testing

We run a diagnostic probe experiment [3] to test whether the topological descriptor H_1 is recoverable from the latent vector z . For each encoder family, we freeze the learned encoder, take latent vectors z , and train a probe, either ridge

Table 11: Latent persistence stability diagnostic at input noise level $\sigma = 0.10$. We add Gaussian noise to the input, re-encode the clean and perturbed sequences, and compare latent-window point clouds and Vietoris–Rips persistence diagrams. d_H is the Hausdorff distance between clean and perturbed latent point clouds, and $d_B(H_k)/d_H$ reports the relative diagram change. Values below one indicate that the persistence diagram changes less than the latent point cloud under the same perturbation.

Dataset	Encoder	$d_H(z, z')$	$d_B(H_0)$	$d_B(H_0)/d_H$	$d_B(H_1)$	$d_B(H_1)/d_H$
Bouncing rings	AE	0.813±0.145	0.470±0.065	0.582±0.037	0.103±0.017	0.128±0.014
	GeoAE	8.946±1.030	1.365±0.121	0.153±0.006	0.317±0.013	0.036±0.006
	TopoAE	0.954±0.057	0.532±0.030	0.558±0.018	0.115±0.007	0.121±0.003
Bouncing disks	AE	0.832±0.127	0.387±0.039	0.468±0.030	0.081±0.005	0.098±0.011
	GeoAE	10.151±1.404	1.660±0.229	0.164±0.011	0.353±0.017	0.035±0.007
	TopoAE	0.797±0.063	0.440±0.043	0.551±0.025	0.095±0.009	0.118±0.005
Orbiting rings	AE	2.562±0.454	1.274±0.119	0.503±0.044	0.376±0.010	0.150±0.028
	GeoAE	10.393±1.153	1.837±0.475	0.175±0.029	0.333±0.014	0.032±0.005
	TopoAE	1.046±0.066	0.535±0.036	0.512±0.014	0.224±0.009	0.215±0.008
Orbiting disks	AE	2.954±0.597	1.344±0.171	0.461±0.040	0.378±0.026	0.131±0.019
	GeoAE	11.621±1.551	2.283±0.717	0.192±0.040	0.372±0.015	0.033±0.006
	TopoAE	0.773±0.062	0.387±0.024	0.501±0.018	0.151±0.007	0.196±0.009
Moving MNIST	AE	2.898±0.385	1.248±0.178	0.432±0.040	0.166±0.027	0.058±0.008
	GeoAE	7.129±0.785	1.156±0.088	0.163±0.015	0.259±0.015	0.037±0.003
	TopoAE	0.780±0.064	0.370±0.028	0.475±0.010	0.070±0.013	0.089±0.010
Lorenz-96	AE	1.599±0.351	0.772±0.231	0.477±0.039	0.008±0.002	0.005±0.000
	GeoAE	12.208±0.784	5.477±0.069	0.450±0.034	0.269±0.020	0.022±0.003
	TopoAE	1.232±0.040	0.522±0.017	0.424±0.011	0.009±0.003	0.007±0.002
ElectricDevices	AE	8.154±0.512	3.315±0.055	0.408±0.025	0.256±0.028	0.031±0.002
	GeoAE	9.817±1.135	3.719±0.122	0.383±0.045	0.331±0.014	0.034±0.003
	TopoAE	7.141±0.413	3.299±0.113	0.463±0.026	0.302±0.026	0.042±0.003
Glioblastoma	AE	16.038±0.174	5.737±1.389	0.357±0.083	0.475±0.093	0.030±0.006
	GeoAE	16.045±0.109	5.507±0.196	0.343±0.013	0.431±0.044	0.027±0.003
	TopoAE	16.181±0.387	5.553±0.393	0.343±0.022	0.654±0.049	0.040±0.003
HeLa	AE	7.967±0.980	4.317±0.322	0.544±0.027	0.099±0.025	0.012±0.002
	GeoAE	15.730±0.149	3.610±0.431	0.230±0.029	0.387±0.018	0.025±0.001
	TopoAE	12.806±3.612	4.206±0.486	0.352±0.106	0.392±0.080	0.032±0.007

regression (linear) or MLP (nonlinear) with 1 hidden layer, to predict the corresponding H_1 topology vector computed from the sliding-window latent point cloud. The probe is trained on the training split and evaluated on held-out test sequences, with R^2 measuring how well z alone predicts the topology feature.

We freeze the encoder and train probes from z to the H_1 topology vector. The linear probe is ridge regression with $\alpha \in \{0.1, 1, 10, 100, 10^3, 10^4, 10^5\}$, selected on a validation split. The nonlinear probe is a one-hidden-layer MLP with hidden size 128, ReLU activation, learning rate 10^{-3} , weight decay 10^{-4} , and a maximum of 300 iterations. For both probes, we use a sequence-level random held-out split with 20% test sequences and 20% validation inside the training set, standardize z and the target topology vector, remove near-constant topology bins with train-set standard deviation $\leq 10^{-3}$, and report held-out R^2 over 5 seeds.

The probe results, which are presented in Table 12, show that for most datasets, the H_1 descriptors are not reliably predictable from z , implying that injecting explicit topology provides information that is not easily accessible from the latent coordinates alone. The exception is Glioblastoma, where R^2 is high for all encoders; ElectricDevices and HeLa also show partial recoverability. This does not contradict our thesis: topology can help either by adding information that is difficult to recover from z , or by making structure that is already partly present in z directly available to the predictor. On several datasets, the nonlinear MLP probe underperforms the linear ridge probe, especially for GeoAE on the motion datasets. This likely reflects overfitting or unstable optimization given limited training samples relative to MLP capacity, rather than proving that nonlinear recoverability is impossible. We therefore treat the MLP probe as redundancy diagnosis.

Table 12: Per-dataset $z \rightarrow H_1$ topology-probe R^2 on held-out examples. Values are mean $\pm$ standard deviation over seeds for both the linear ridge and nonlinear MLP probes.

Dataset	Linear ridge probe			Nonlinear MLP probe		
	AE	GeoAE	TopoAE	AE	GeoAE	TopoAE
Bouncing disks	0.035 $\pm$ 0.011	0.006 $\pm$ 0.006	0.026 $\pm$ 0.007	0.044 $\pm$ 0.018	-0.027 $\pm$ 0.022	0.038 $\pm$ 0.011
Bouncing rings	0.024 $\pm$ 0.013	-0.000 $\pm$ 0.001	0.036 $\pm$ 0.013	0.034 $\pm$ 0.018	-0.031 $\pm$ 0.011	0.045 $\pm$ 0.017
Orbiting disks	0.006 $\pm$ 0.004	0.004 $\pm$ 0.006	0.020 $\pm$ 0.013	0.032 $\pm$ 0.012	-0.022 $\pm$ 0.016	0.065 $\pm$ 0.013
Orbiting rings	-0.003 $\pm$ 0.005	-0.002 $\pm$ 0.004	0.013 $\pm$ 0.011	-0.008 $\pm$ 0.008	-0.031 $\pm$ 0.012	0.033 $\pm$ 0.014
Moving MNIST	-0.001 $\pm$ 0.010	-0.006 $\pm$ 0.007	0.014 $\pm$ 0.012	-0.006 $\pm$ 0.028	-0.085 $\pm$ 0.056	0.016 $\pm$ 0.027
Lorenz-96	-0.003 $\pm$ 0.002	-0.003 $\pm$ 0.005	0.000 $\pm$ 0.002	-0.024 $\pm$ 0.016	-0.171 $\pm$ 0.187	-0.013 $\pm$ 0.008
ElectricDevices	0.070 $\pm$ 0.054	0.229 $\pm$ 0.061	0.117 $\pm$ 0.046	0.125 $\pm$ 0.052	0.311 $\pm$ 0.062	0.224 $\pm$ 0.076
Glioblastoma	0.775 $\pm$ 0.050	0.764 $\pm$ 0.053	0.770 $\pm$ 0.021	0.958 $\pm$ 0.030	0.944 $\pm$ 0.113	0.870 $\pm$ 0.146
HeLa	0.048 $\pm$ 0.700	0.427 $\pm$ 0.143	0.427 $\pm$ 0.119	0.040 $\pm$ 0.363	0.581 $\pm$ 0.348	0.404 $\pm$ 0.381

B Other Scenarios

This part includes additional ideas we explored.

B.1 3D scenario

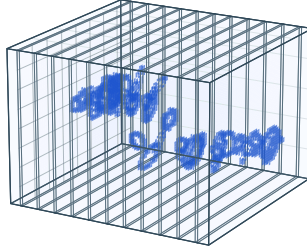


Figure 6: Visualization of the 3D construction of the Bouncing Rings dataset. Consecutive frames are stacked along the time axis to form a cubical volume, on which we compute H_0 , H_1 , and H_2 persistence features

We tested a spatiotemporal TDA variant, shown in Figure 6, where a sequence X is treated as a 3D volume with axes (t, y, x) . Instead of computing the persistence on a latent point cloud, we stack consecutive input frames along the time axis and compute cubical persistence [29] directly on the resulting volume. For time series datasets, sequence values are arranged into 2D frame-like arrays, which are then stacked over time to form the 3D cubical volume. We then extract Betti curve features from H_0 , H_1 , and H_2 , and their combinations, and concatenate the selected features with the latents z for prediction. In this setting, H_0 tracks connected spatiotemporal components, H_1 tracks loop structures, and H_2 tracks enclosed voids or cavities in the volume.

We run this algorithm once per dataset, with results shown in Table 13. The best 3D TDA mode improves weighted MSE over the no-TDA baseline on all datasets, suggesting that volume-level topology can provide useful auxiliary signal. However, because this method computes cubical persistence in input space X , it is substantially slower, taking several hours to produce Table 13. We treat this as an exploratory extension, rather than a main method, since the latent point cloud approach gives similar gains at much lower computational cost. Its consistent improvement across datasets suggests that spatiotemporal H_2 and other persistence features may be useful in settings where the extra computational cost is acceptable.

Table 13: Exploratory 3D video TDA results. Frames are treated as a spatiotemporal volume and cubical persistence features are appended for prediction. Each row reports a single run, comparing the no-TDA baseline against the best 3D-TDA mode by weighted MSE. Lower MSE is better.

Dataset	No TDA	Best 3D-TDA	Best mode	Δ	Pixel MSE
Bouncing rings	0.1432	0.1323	$H_0 + H_1$	+7.6%	0.0796
Bouncing disks	0.1509	0.1402	$H_0 + H_1$	+7.1%	0.0711
Orbiting rings	0.1506	0.1488	$H_0 + H_2$	+1.2%	0.0677
Orbiting disks	0.1765	0.1751	H_0	+0.8%	0.0971
Moving MNIST	0.2269	0.2264	$H_0 + H_1 + H_2$	+0.2%	0.0987
Glioblastoma	0.0220	0.0217	H_0	+1.4%	0.0020
HeLa	0.1410	0.1202	$H_0 + H_2$	+14.8%	0.0112
Lorenz-96	0.0233	0.0227	H_2	+2.6%	0.0130
ElectricDevices	0.0477	0.0474	$H_0 + H_1 + H_2$	+0.6%	0.0304

B.2 V-JEPA

We also evaluate whether the topological signal helps when the latent representation comes from a large pretrained video encoder. We use the V-JEPA2 model ‘facebook/vjepa2-vit1-fpc64-256’ [link], and train only the downstream predictor and fusion head on top of its latent features. Because V-JEPA is computationally expensive, we run this robustness check for a single run with `latent_tda_max_train` = 64 and `latent_tda_max_test` = 32.

Table 14: Single-run frozen V-JEPA robustness check. V-JEPA is frozen and only the temporal predictor/fusion head is trained. We cap the run at 64 training sequences and 32 test sequences. Lower MSE is better; Gain is the relative MSE reduction of the best topology-fusion mode over raw V-JEPA z .

Dataset	V-JEPA z MSE	Best topo mode	Best topo MSE	Best topo R^2	Gain
Bouncing rings	0.0228	H1	0.0220	0.9780	3.5%
Bouncing disks	0.0215	PI-H1	0.0193	0.9806	10.2%
Orbiting rings	0.0294	PersLay-H1	0.0272	0.9726	7.5%
Orbiting disks	0.0301	PI-H1	0.0288	0.9711	4.3%
Moving MNIST	0.0394	PersLay-H1	0.0371	0.9629	5.8%
Glioblastoma	0.0290	PersLay-H1	0.0289	0.9708	0.3%
HeLa	0.0632	PersLay-H1	0.0629	0.9358	0.5%
Lorenz-96	0.0116	PI-H1	0.0095	0.9905	18.1%
ElectricDevices	0.0430	PersLay-H1	0.0480	0.9526	-11.6%

Results, presented in Table 14, show that adding topology improves frozen V-JEPA2 forecasting on 8/9 datasets, with an average MSE reduction of roughly 4.3% over raw V-JEPA2 latents. We treat this as a robustness check rather than a main benchmark: V-JEPA2 is substantially heavier than our dataset-specific encoders and is unnecessary for our low-dimensional forecasting setting, but the result suggests that topological auxiliary features can improve latent forecasting even when the representation comes from a large frozen pretrained video model.

C Implementation Details

C.1 Hardware

We ran the experiments on a workstation with an Intel Core i7-13700 CPU, 30 GB RAM, and an NVIDIA GeForce RTX 4090 (24 GB VRAM) running Ubuntu 24.10 with CUDA 13.0 and PyTorch 2.13.0. We also used an Apple MacBook Pro with an M4 Pro (14-core CPU) chip, 48 GB unified memory, running macOS 26.5.2 and PyTorch 2.8.0. We compute Vietoris–Rips (VR) persistence of latent point clouds using Ripser [10], persistence image and persistence landscape vectorizations using GUDHI [48], and bottleneck distances in the stability diagnostic using Persim [43].

C.2 Model Architecture

All autoencoder variants use the same CNN encoder architecture, differing only in training objective. The encoder maps each grayscale frame $X_t \in \mathbb{R}^{1 \times H \times W}$ to a latent vector z_t , using two convolutional layers, $1 \rightarrow 16$ and $16 \rightarrow 32$,

both with kernel size 5, stride 2, padding 2, and ReLU activation. The resulting feature map is reduced using adaptive average pooling to 4×4 , flattened, and projected with a linear layer to the dataset-specific latent dimension. We use latent dimension 48 for Moving MNIST, 64 for the synthetic video datasets, Lorenz-96, and ElectricDevices, and 128 for HeLa and Glioblastoma.

Unless otherwise stated, reconstruction uses an MLP decoder with Linear($d_z, 512$), ReLU, Linear($512, HW$), and sigmoid output. For forecasting, we use a one-layer LSTM followed by a linear projection to the latent dimension. The predictor hidden dimension is selected by validation on the z -only baseline and then fixed for all modes: 128 for Lorenz-96 and 256 for the remaining datasets. In learned-fusion modes, $[z_t, B_t]$ is passed through a fusion MLP with Linear–GELU–LayerNorm–Linear–GELU, producing a fused vector before the LSTM predictor.

For each dataset, we performed a small validation-based hyperparameter search using only the z -only latent forecasting baseline. We varied the predictor learning rate over $[0.1, 0.3, 1, 3, 5, 10] \times 10^{-3}$, hidden dimension over $[64, 128, 256]$, and training epochs over $[8, 20, 40]$, selecting the configuration with the lowest validation MSE. The selected dataset-level configuration was then fixed and reused for all topology, PCA, KPCA, and fusion variants, so that improvements are not due to mode-specific hyperparameter tuning.

D Correlations and Runs

D.1 PCA

Figure 7 reports z -only forecasting MSE across prediction horizons. Error generally increases with horizon size.

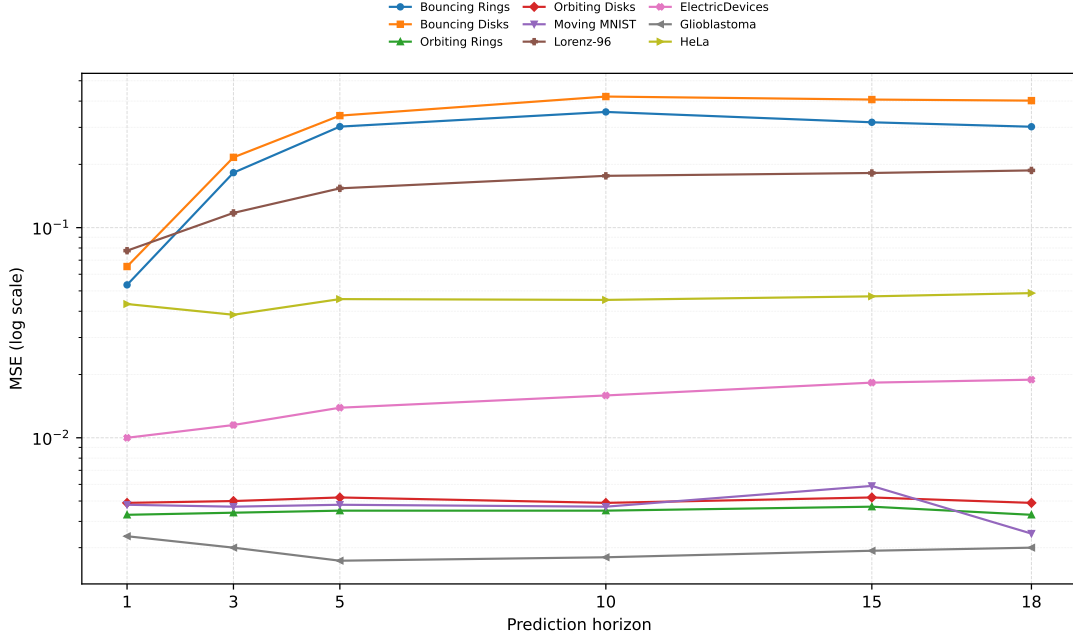


Figure 7: Prediction MSE vs varied prediction horizon values for all datasets

We start our analysis with the explained variance per each PCA component. Figure 8 shows the cumulative variance as a function of the number of principal components, noting that GeoAE has the most evenly-spread PCA explainability as it grows the slowest.

Figure 9 shows the PCA effective rank as a function of geometry regularization strength λ_{GeoAE} . Effective rank generally increases from $\lambda = 0$, indicating that geometry regularization prevents latent variance from concentrating in only few principal directions. For most datasets, it peaks around $\lambda = 0.1 - 0.2$ and subsequently decreases. GeoAE encourages the preservation of normalized pairwise distances, rather than forcing latent coordinates to reproduce the input. Strong regularization could constrain the encoder and reduce its ability to balance geometry preservation with reconstruction. So, intermediate λ values often produce the most distributed latent representations, though a high effective rank on its own does not imply optimal forecasting performance.

Next Figure 10 shows prediction MSE as function of geometric regularization. Across all datasets, $\lambda_{\text{GeoAE}} = 0$ produces the highest MSE, while the smallest regularization strengths substantially reduce MSE. Because the reconstruction and geometric losses may have different numerical scales, a small λ_{GeoAE} value does not necessarily imply a small

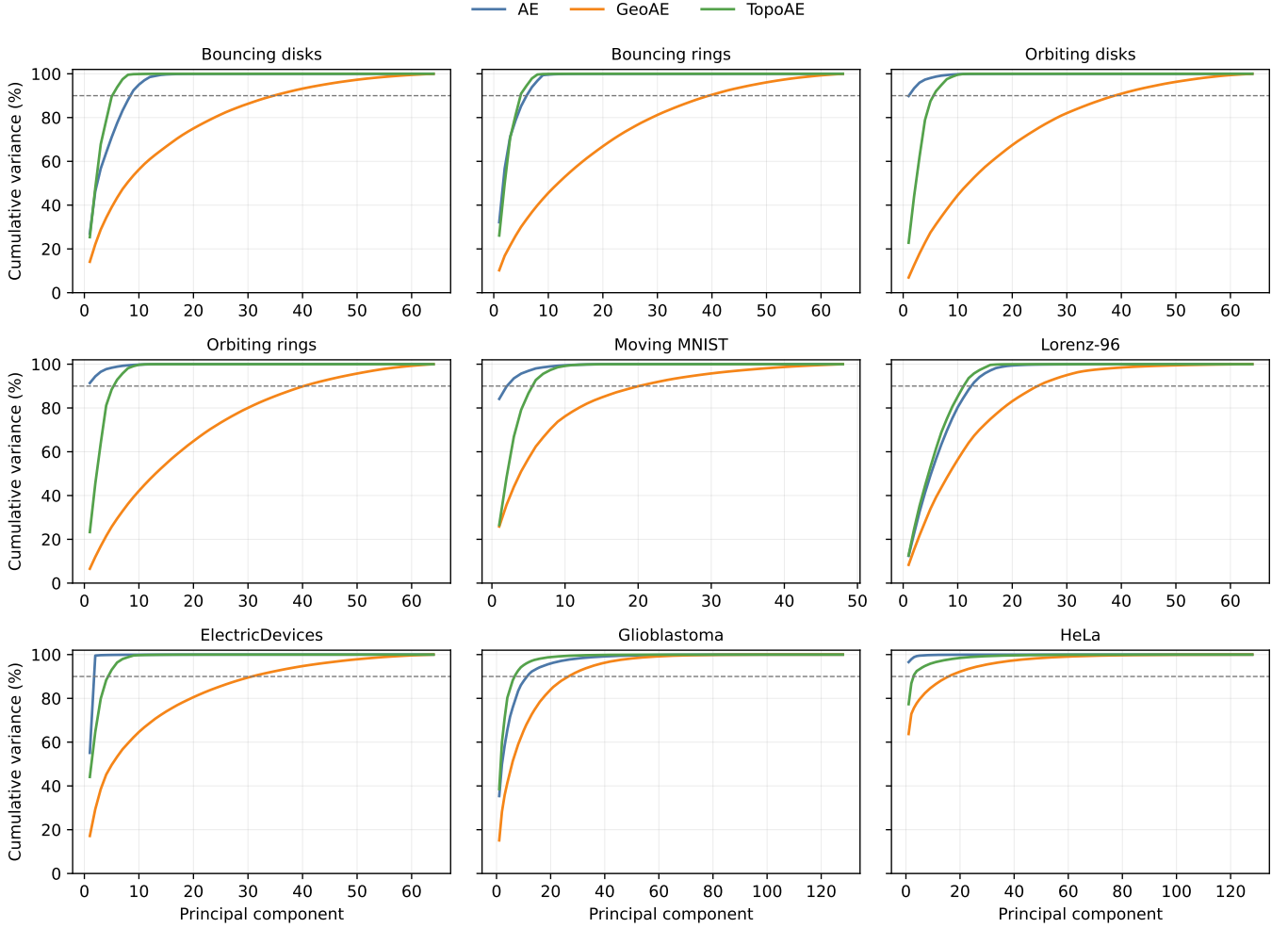


Figure 8: Cumulative PCA variance for each dataset

contribution to the total loss. Performance then plateaus, with the optimal λ varying across datasets. This implies that preserving input-space distances produces a more structured latent representation for prediction, though stronger regularization is not always beneficial as it may restrict the encoder’s ability to retain task-relevant features.

Next, Figure 11 shows prediction MSE against the PCA effective rank of (z) for each dataset. Generally, higher effective rank is associated with lower MSE, suggesting that prediction improves when information is spread across more latent dimensions. As shown in Figures 9 and 10, increasing λ_{GeoAE} initially increases effective rank and reduces MSE. However, very strong regularization can reduce the encoder’s flexibility, causing the rank to decrease.

The lowest MSE does not always occur at the highest effective rank. Effective rank measures how widely information is spread, but not the usefulness of that information for prediction. Some latent dimensions may capture reconstruction details or noise rather than predictive dynamics. Therefore, a well-distributed latent space can help forecasting, but maximizing effective rank alone does not guarantee the best prediction.

Figure 12 shows the PCA projections for each dataset and encoder type, showing the explained variance by PC1 and PC2. As this figure only shows the components on 2 axes, where different axes not shown here may exhibit different behaviours, it is shown for demonstration purposes only.

PCA effective rank versus λ_{GeoAE} , by dataset

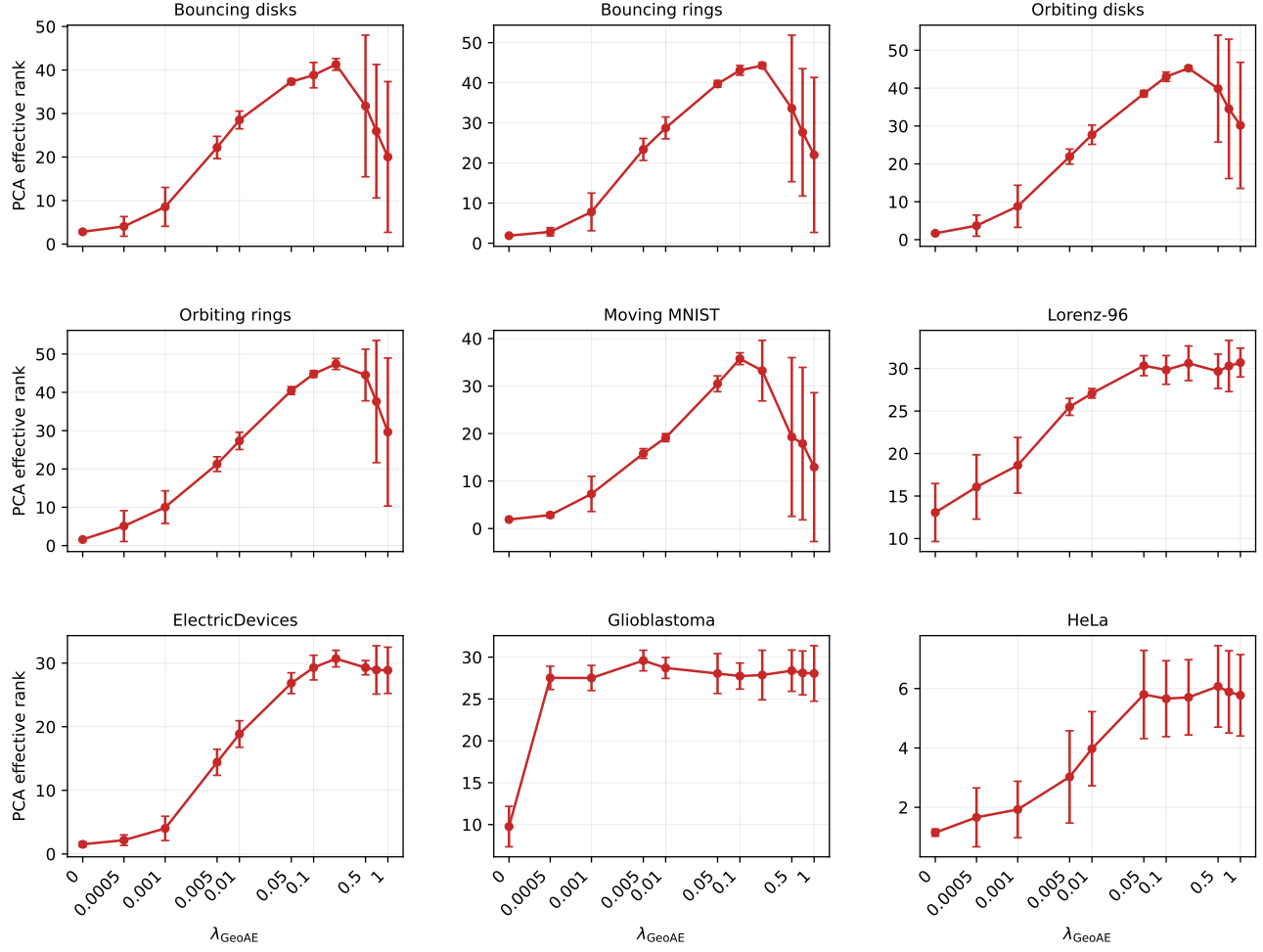


Figure 9: PCA rank vs λ_{GeoAE} , per dataset

D.2 Additional Runs

Table 15 shows the results for all datasets, scenarios, and encoders.

Table 15: Complete mode-level forecasting results in a compact encoder-grouped layout. Entries are five-seed mean $\pm$ SD. Lower MSE and higher R^2 are better.

Dataset	Mode	AE		GeoAE		TopoAE	
		MSE	R^2	MSE	R^2	MSE	R^2
Bouncing disks	z	0.3429 $\pm$ 0.0266	0.6586 $\pm$ 0.0260	0.0013 $\pm$ 0.0003	0.9987 $\pm$ 0.0003	0.2158 $\pm$ 0.0099	0.7868 $\pm$ 0.0098
	topo.h1	0.5289 $\pm$ 0.0446	0.4734 $\pm$ 0.0435	0.0088 $\pm$ 0.0016	0.9912 $\pm$ 0.0016	0.2943 $\pm$ 0.0148	0.7093 $\pm$ 0.0146
	topo.pi.h1	0.6041 $\pm$ 0.0714	0.3985 $\pm$ 0.0707	0.0082 $\pm$ 0.0012	0.9918 $\pm$ 0.0012	0.3075 $\pm$ 0.0142	0.6963 $\pm$ 0.0141
	z.both	0.3697 $\pm$ 0.0291	0.6319 $\pm$ 0.0285	0.0014 $\pm$ 0.0002	0.9986 $\pm$ 0.0002	0.2155 $\pm$ 0.0086	0.7872 $\pm$ 0.0085
	z.diffusion.h1	0.3453 $\pm$ 0.0291	0.6562 $\pm$ 0.0285	0.0015 $\pm$ 0.0003	0.9985 $\pm$ 0.0003	0.2166 $\pm$ 0.0083	0.7861 $\pm$ 0.0082
	z.fuse.diffusion.h1	0.3670 $\pm$ 0.0297	0.6346 $\pm$ 0.0289	0.0023 $\pm$ 0.0004	0.9977 $\pm$ 0.0004	0.2235 $\pm$ 0.0092	0.7792 $\pm$ 0.0091
	z.fuse.h1	0.3639 $\pm$ 0.0297	0.6377 $\pm$ 0.0290	0.0025 $\pm$ 0.0003	0.9975 $\pm$ 0.0003	0.2259 $\pm$ 0.0075	0.7769 $\pm$ 0.0074
	z.fuse.kpca.h1	0.3644 $\pm$ 0.0277	0.6372 $\pm$ 0.0270	0.0025 $\pm$ 0.0003	0.9975 $\pm$ 0.0003	0.2217 $\pm$ 0.0096	0.7811 $\pm$ 0.0095
	z.fuse.landscape.h1	0.3682 $\pm$ 0.0296	0.6334 $\pm$ 0.0288	0.0026 $\pm$ 0.0003	0.9974 $\pm$ 0.0003	0.2356 $\pm$ 0.0108	0.7674 $\pm$ 0.0107
	z.fuse.laplacian.h1	0.3634 $\pm$ 0.0283	0.6382 $\pm$ 0.0275	0.0021 $\pm$ 0.0003	0.9979 $\pm$ 0.0003	0.2286 $\pm$ 0.0104	0.7743 $\pm$ 0.0103
	z.fuse.pca.h1	0.4270 $\pm$ 0.0273	0.5748 $\pm$ 0.0267	0.0022 $\pm$ 0.0007	0.9978 $\pm$ 0.0007	0.2258 $\pm$ 0.0100	0.7770 $\pm$ 0.0099
	z.fuse.perslay.h1	0.3671 $\pm$ 0.0259	0.6345 $\pm$ 0.0252	0.0019 $\pm$ 0.0007	0.9981 $\pm$ 0.0007	0.2293 $\pm$ 0.0081	0.7735 $\pm$ 0.0080
	z.fuse.pi.h1	0.3656 $\pm$ 0.0293	0.6359 $\pm$ 0.0286	0.0026 $\pm$ 0.0003	0.9974 $\pm$ 0.0003	0.2263 $\pm$ 0.0120	0.7765 $\pm$ 0.0119
	z.fuse.rff.h1	0.3672 $\pm$ 0.0283	0.6344 $\pm$ 0.0273	0.0024 $\pm$ 0.0005	0.9976 $\pm$ 0.0005	0.2267 $\pm$ 0.0069	0.7761 $\pm$ 0.0068
	z.h0	0.3764 $\pm$ 0.0337	0.6252 $\pm$ 0.0330	0.0012 $\pm$ 0.0002	0.9988 $\pm$ 0.0002	0.2163 $\pm$ 0.0096	0.7863 $\pm$ 0.0095
	z.h1	0.3402 $\pm$ 0.0256	0.6613 $\pm$ 0.0251	0.0013 $\pm$ 0.0003	0.9987 $\pm$ 0.0003	0.2142 $\pm$ 0.0091	0.7884 $\pm$ 0.0090
	z.landscape.both	0.3531 $\pm$ 0.0289	0.6484 $\pm$ 0.0282	0.0011 $\pm$ 0.0002	0.9989 $\pm$ 0.0002	0.2070 $\pm$ 0.0083	0.7956 $\pm$ 0.0082
	z.landscape.h0	0.3553 $\pm$ 0.0309	0.6463 $\pm$ 0.0301	0.0010 $\pm$ 0.0002	0.9990 $\pm$ 0.0002	0.2098 $\pm$ 0.0093	0.7928 $\pm$ 0.0092
	z.landscape.h1	0.3436 $\pm$ 0.0259	0.6578 $\pm$ 0.0253	0.0015 $\pm$ 0.0002	0.9985 $\pm$ 0.0002	0.2142 $\pm$ 0.0093	0.7885 $\pm$ 0.0092
	z.laplacian.h1	0.3461 $\pm$ 0.0276	0.6553 $\pm$ 0.0270	0.0014 $\pm$ 0.0002	0.9986 $\pm$ 0.0002	0.2152 $\pm$ 0.0079	0.7875 $\pm$ 0.0079
	z.perslay.both	0.3778 $\pm$ 0.0288	0.6238 $\pm$ 0.0281	0.0011 $\pm$ 0.0001	0.9989 $\pm$ 0.0001	0.2117 $\pm$ 0.0078	0.7909 $\pm$ 0.0077
	z.perslay.h0	0.3909 $\pm$ 0.0345	0.6108 $\pm$ 0.0337	0.0011 $\pm$ 0.0001	0.9989 $\pm$ 0.0001	0.2114 $\pm$ 0.0080	0.7913 $\pm$ 0.0079
	z.perslay.h1	0.3449 $\pm$ 0.0307	0.6566 $\pm$ 0.0299	0.0015 $\pm$ 0.0003	0.9985 $\pm$ 0.0003	0.2156 $\pm$ 0.0102	0.7871 $\pm$ 0.0101

Continued on next page

Table 15 – continued

Dataset	Mode	AE		GeoAE		TopoAE		
		MSE	R ²	MSE	R ²	MSE	R ²	
	z.pi.both	0.4732 ± 0.0394	0.5289 ± 0.0384	0.0013 ± 0.0002	0.9987 ± 0.0002	0.2197 ± 0.0091	0.7830 ± 0.0090	
	z.pi.h0	0.4977 ± 0.0434	0.5044 ± 0.0423	0.0012 ± 0.0001	0.9988 ± 0.0001	0.2163 ± 0.0101	0.7864 ± 0.0100	
	z.pi.h1	0.3461 ± 0.0290	0.6554 ± 0.0283	0.0015 ± 0.0001	0.9985 ± 0.0001	0.2143 ± 0.0094	0.7884 ± 0.0093	
	z.rff.h1	0.3437 ± 0.0282	0.6578 ± 0.0276	0.0014 ± 0.0003	0.9986 ± 0.0003	0.2177 ± 0.0091	0.7850 ± 0.0090	
Bouncing rings	z	0.2972 ± 0.0474	0.7035 ± 0.0474	0.0013 ± 0.0001	0.9987 ± 0.0001	0.2104 ± 0.0120	0.7928 ± 0.0117	
	topo.h1	0.4238 ± 0.0867	0.5772 ± 0.0867	0.0091 ± 0.0014	0.9909 ± 0.0014	0.2701 ± 0.0151	0.7339 ± 0.0147	
	topo.pi.h1	0.4883 ± 0.1033	0.5129 ± 0.1031	0.0080 ± 0.0006	0.9920 ± 0.0006	0.2902 ± 0.0186	0.7141 ± 0.0181	
	z.both	0.3232 ± 0.0568	0.6776 ± 0.0568	0.0014 ± 0.0002	0.9986 ± 0.0002	0.2072 ± 0.0110	0.7959 ± 0.0108	
	z.diffusion.h1	0.3002 ± 0.0486	0.7006 ± 0.0486	0.0014 ± 0.0003	0.9986 ± 0.0003	0.2098 ± 0.0113	0.7934 ± 0.0110	
	z.fuse.diffusion.h1	0.3142 ± 0.0568	0.6865 ± 0.0568	0.0023 ± 0.0002	0.9977 ± 0.0002	0.2142 ± 0.0116	0.7890 ± 0.0113	
	z.fuse.h1	0.3112 ± 0.0561	0.6895 ± 0.0561	0.0023 ± 0.0002	0.9977 ± 0.0002	0.2160 ± 0.0105	0.7872 ± 0.0102	
	z.fuse.kpca.h1	0.3104 ± 0.0557	0.6904 ± 0.0557	0.0023 ± 0.0003	0.9977 ± 0.0003	0.2148 ± 0.0132	0.7884 ± 0.0129	
	z.fuse.landscape.h1	0.3189 ± 0.0522	0.6819 ± 0.0521	0.0027 ± 0.0004	0.9973 ± 0.0004	0.2255 ± 0.0122	0.7778 ± 0.0121	
	z.fuse.laplacian.h1	0.3152 ± 0.0554	0.6856 ± 0.0553	0.0022 ± 0.0002	0.9978 ± 0.0002	0.2160 ± 0.0115	0.7873 ± 0.0112	
	z.fuse.pca.h1	0.3902 ± 0.0620	0.6108 ± 0.0621	0.0022 ± 0.0003	0.9978 ± 0.0003	0.2187 ± 0.0153	0.7846 ± 0.0150	
	z.fuse.perslay.h1	0.3189 ± 0.0573	0.6819 ± 0.0573	0.0023 ± 0.0002	0.9977 ± 0.0002	0.2175 ± 0.0113	0.7857 ± 0.0110	
	z.fuse.pi.h1	0.3162 ± 0.0546	0.6846 ± 0.0546	0.0022 ± 0.0002	0.9978 ± 0.0002	0.2168 ± 0.0126	0.7864 ± 0.0124	
	z.fuse.rff.h1	0.3116 ± 0.0532	0.6892 ± 0.0532	0.0019 ± 0.0005	0.9981 ± 0.0005	0.2208 ± 0.0089	0.7825 ± 0.0086	
	z.h0	0.3281 ± 0.0622	0.6727 ± 0.0622	0.0012 ± 0.0002	0.9988 ± 0.0002	0.2086 ± 0.0105	0.7945 ± 0.0102	
	z.h1	0.2967 ± 0.0480	0.7040 ± 0.0480	0.0013 ± 0.0002	0.9987 ± 0.0002	0.2104 ± 0.0103	0.7927 ± 0.0101	
	z.landscape.both	0.3018 ± 0.0526	0.6989 ± 0.0526	0.0013 ± 0.0002	0.9987 ± 0.0002	0.2057 ± 0.0107	0.7973 ± 0.0104	
	z.landscape.h0	0.3028 ± 0.0524	0.6980 ± 0.0524	0.0011 ± 0.0002	0.9989 ± 0.0002	0.2076 ± 0.0107	0.7954 ± 0.0105	
	z.landscape.h1	0.2955 ± 0.0495	0.7052 ± 0.0496	0.0014 ± 0.0003	0.9986 ± 0.0003	0.2085 ± 0.0125	0.7946 ± 0.0122	
	z.laplacian.h1	0.2986 ± 0.0509	0.7022 ± 0.0509	0.0013 ± 0.0003	0.9987 ± 0.0003	0.2096 ± 0.0091	0.7935 ± 0.0089	
	z.perslay.both	0.3320 ± 0.0587	0.6688 ± 0.0587	0.0012 ± 0.0001	0.9988 ± 0.0001	0.2035 ± 0.0132	0.7996 ± 0.0129	
	z.perslay.h0	0.3377 ± 0.0583	0.6631 ± 0.0583	0.0011 ± 0.0001	0.9989 ± 0.0001	0.2024 ± 0.0103	0.8006 ± 0.0101	
	z.perslay.h1	0.2976 ± 0.0482	0.7032 ± 0.0482	0.0013 ± 0.0003	0.9987 ± 0.0003	0.2082 ± 0.0115	0.7948 ± 0.0112	
	z.pi.both	0.4064 ± 0.0795	0.5946 ± 0.0795	0.0012 ± 0.0001	0.9988 ± 0.0001	0.2111 ± 0.0103	0.7920 ± 0.0100	
	z.pi.h0	0.4084 ± 0.0811	0.5926 ± 0.0811	0.0012 ± 0.0002	0.9988 ± 0.0002	0.2106 ± 0.0106	0.7925 ± 0.0104	
	z.pi.h1	0.2989 ± 0.0479	0.7019 ± 0.0479	0.0014 ± 0.0004	0.9986 ± 0.0004	0.2098 ± 0.0110	0.7933 ± 0.0107	
	z.rff.h1	0.2937 ± 0.0467	0.7070 ± 0.0467	0.0013 ± 0.0002	0.9987 ± 0.0002	0.2082 ± 0.0121	0.7949 ± 0.0118	
	Orbiting disks	z	0.0052 ± 0.0018	0.9949 ± 0.0017	0.0017 ± 0.0001	0.9983 ± 0.0001	0.2281 ± 0.0124	0.7810 ± 0.0117
		topo.h1	0.0190 ± 0.0031	0.9813 ± 0.0030	0.0126 ± 0.0019	0.9874 ± 0.0019	0.3611 ± 0.0251	0.6531 ± 0.0241
		topo.pi.h1	0.0207 ± 0.0044	0.9796 ± 0.0043	0.0105 ± 0.0009	0.9895 ± 0.0009	0.3633 ± 0.0250	0.6511 ± 0.0240
		z.both	0.0053 ± 0.0017	0.9948 ± 0.0017	0.0019 ± 0.0002	0.9981 ± 0.0002	0.2262 ± 0.0154	0.7828 ± 0.0146
z.diffusion.h1		0.0051 ± 0.0017	0.9950 ± 0.0017	0.0018 ± 0.0002	0.9982 ± 0.0002	0.2235 ± 0.0107	0.7854 ± 0.0102	
z.fuse.diffusion.h1		0.0071 ± 0.0028	0.9931 ± 0.0027	0.0017 ± 0.0002	0.9983 ± 0.0002	0.2761 ± 0.0148	0.7348 ± 0.0144	
z.fuse.h1		0.0070 ± 0.0028	0.9931 ± 0.0027	0.0015 ± 0.0001	0.9985 ± 0.0001	0.2761 ± 0.0162	0.7348 ± 0.0153	
z.fuse.kpca.h1		0.0069 ± 0.0027	0.9932 ± 0.0027	0.0016 ± 0.0001	0.9984 ± 0.0001	0.2800 ± 0.0105	0.7311 ± 0.0097	
z.fuse.landscape.h1		0.0068 ± 0.0027	0.9933 ± 0.0027	0.0015 ± 0.0001	0.9985 ± 0.0001	0.2851 ± 0.0132	0.7262 ± 0.0127	
z.fuse.laplacian.h1		0.0069 ± 0.0027	0.9932 ± 0.0027	0.0017 ± 0.0001	0.9983 ± 0.0001	0.2818 ± 0.0153	0.7294 ± 0.0145	
z.fuse.pca.h1		0.0064 ± 0.0023	0.9937 ± 0.0023	0.0016 ± 0.0002	0.9984 ± 0.0002	0.2766 ± 0.0071	0.7343 ± 0.0067	
z.fuse.perslay.h1		0.0069 ± 0.0027	0.9932 ± 0.0026	0.0017 ± 0.0001	0.9983 ± 0.0001	0.2869 ± 0.0210	0.7244 ± 0.0202	
z.fuse.pi.h1		0.0069 ± 0.0029	0.9932 ± 0.0028	0.0016 ± 0.0001	0.9984 ± 0.0001	0.2857 ± 0.0172	0.7256 ± 0.0164	
z.fuse.rff.h1		0.0069 ± 0.0028	0.9932 ± 0.0027	0.0018 ± 0.0001	0.9982 ± 0.0001	0.2635 ± 0.0213	0.7469 ± 0.0204	
z.h0		0.0055 ± 0.0014	0.9946 ± 0.0014	0.0019 ± 0.0002	0.9981 ± 0.0002	0.2215 ± 0.0105	0.7872 ± 0.0100	
z.h1		0.0051 ± 0.0016	0.9950 ± 0.0015	0.0016 ± 0.0001	0.9984 ± 0.0001	0.2209 ± 0.0090	0.7879 ± 0.0087	
z.landscape.both		0.0064 ± 0.0019	0.9937 ± 0.0018	0.0017 ± 0.0001	0.9983 ± 0.0001	0.2112 ± 0.0076	0.7971 ± 0.0072	
z.landscape.h0		0.0059 ± 0.0022	0.9942 ± 0.0021	0.0014 ± 0.0002	0.9986 ± 0.0002	0.2120 ± 0.0102	0.7964 ± 0.0097	
z.landscape.h1		0.0052 ± 0.0017	0.9949 ± 0.0017	0.0018 ± 0.0003	0.9982 ± 0.0003	0.2083 ± 0.0076	0.7999 ± 0.0073	
z.laplacian.h1		0.0051 ± 0.0016	0.9950 ± 0.0015	0.0019 ± 0.0001	0.9981 ± 0.0001	0.2280 ± 0.0095	0.7811 ± 0.0089	
z.perslay.both		0.0055 ± 0.0019	0.9946 ± 0.0019	0.0013 ± 0.0001	0.9987 ± 0.0001	0.2256 ± 0.0046	0.7833 ± 0.0044	
z.perslay.h0		0.0047 ± 0.0016	0.9954 ± 0.0015	0.0011 ± 0.0001	0.9989 ± 0.0001	0.2184 ± 0.0047	0.7903 ± 0.0042	
z.perslay.h1		0.0054 ± 0.0018	0.9947 ± 0.0017	0.0017 ± 0.0000	0.9983 ± 0.0000	0.2157 ± 0.0116	0.7928 ± 0.0111	
z.pi.both		0.0066 ± 0.0021	0.9935 ± 0.0021	0.0021 ± 0.0002	0.9979 ± 0.0002	0.2637 ± 0.0080	0.7467 ± 0.0080	
z.pi.h0		0.0072 ± 0.0019	0.9929 ± 0.0018	0.0025 ± 0.0002	0.9975 ± 0.0002	0.2552 ± 0.0098	0.7549 ± 0.0094	
z.pi.h1		0.0051 ± 0.0015	0.9950 ± 0.0014	0.0018 ± 0.0001	0.9982 ± 0.0001	0.2208 ± 0.0073	0.7879 ± 0.0071	
z.rff.h1		0.0051 ± 0.0015	0.9950 ± 0.0015	0.0017 ± 0.0001	0.9984 ± 0.0001	0.2239 ± 0.0079	0.7849 ± 0.0076	
Orbiting rings	z	0.0045 ± 0.0014	0.9956 ± 0.0014	0.0017 ± 0.0001	0.9983 ± 0.0001	0.2321 ± 0.0144	0.7788 ± 0.0133	
	topo.h1	0.0169 ± 0.0032	0.9834 ± 0.0031	0.0132 ± 0.0012	0.9868 ± 0.0012	0.3221 ± 0.0380	0.6929 ± 0.0356	
	topo.pi.h1	0.0236 ± 0.0086	0.9768 ± 0.0086	0.0109 ± 0.0012	0.9891 ± 0.0012	0.3232 ± 0.0386	0.6920 ± 0.0361	
	z.both	0.0045 ± 0.0013	0.9956 ± 0.0013	0.0020 ± 0.0002	0.9980 ± 0.0002	0.2337 ± 0.0091	0.7772 ± 0.0086	
	z.diffusion.h1	0.0046 ± 0.0014	0.9955 ± 0.0014	0.0018 ± 0.0001	0.9982 ± 0.0001	0.2394 ± 0.0141	0.7718 ± 0.0130	
	z.fuse.diffusion.h1	0.0055 ± 0.0020	0.9946 ± 0.0020	0.0016 ± 0.0001	0.9984 ± 0.0001	0.2714 ± 0.0259	0.7412 ± 0.0242	
	z.fuse.h1	0.0056 ± 0.0019	0.9945 ± 0.0019	0.0017 ± 0.0001	0.9983 ± 0.0001	0.2617 ± 0.0202	0.7505 ± 0.0189	
	z.fuse.kpca.h1	0.0056 ± 0.0021	0.9945 ± 0.0020	0.0016 ± 0.0001	0.9984 ± 0.0001	0.2653 ± 0.0232	0.7471 ± 0.0217	
	z.fuse.landscape.h1	0.0055 ± 0.0021	0.9946 ± 0.0020	0.0017 ± 0.0001	0.9983 ± 0.0001	0.2796 ± 0.0298	0.7335 ± 0.0279	
	z.fuse.laplacian.h1	0.0056 ± 0.0020	0.9945 ± 0.0020	0.0016 ± 0.0001	0.9984 ± 0.0001	0.2669 ± 0.0212	0.7456 ± 0.0197	
	z.fuse.pca.h1	0.0055 ± 0.0019	0.9946 ± 0.0018	0.0014 ± 0.0001	0.9986 ± 0.0001	0.2723 ± 0.0302	0.7405 ± 0.0282	
	z.fuse.perslay.h1	0.0054 ± 0.0020	0.9947 ± 0.0020	0.0016 ± 0.0002	0.9984 ± 0.0002	0.2714 ± 0.0253	0.7412 ± 0.0236	
	z.fuse.pi.h1	0.0057 ± 0.0020	0.9944 ± 0.0019	0.0017 ± 0.0001	0.9983 ± 0.0001	0.2712 ± 0.0305	0.7415 ± 0.0285	
	z.fuse.rff.h1	0.0055 ± 0.0019	0.9946 ± 0.0019	0.0016 ± 0.0001	0.9984 ± 0.0001	0.2688 ± 0.0255	0.7437 ± 0.0238	
	z.h0	0.0046 ± 0.0014	0.9955 ± 0.0013	0.0020 ± 0.0001	0.9980 ± 0.0001	0.2291 ± 0.0062	0.7816 ± 0.0058	
	z.h1	0.0047 ± 0.0016	0.9954 ± 0.0016	0.0019 ± 0.0002	0.9981 ± 0.0002	0.2279 ± 0.0143	0.7827 ± 0.0135	
	z.landscape.both	0.0055 ± 0.0015	0.9946 ± 0.0015	0.0018 ± 0.0003	0.9982 ± 0.0003	0.2282 ± 0.0086	0.7824 ± 0.0080	
	z.landscape.h0	0.0049 ± 0.0014	0.9952 ± 0.0014	0.0012 ± 0.0002	0.9988 ± 0.0002	0.2293 ± 0.0021	0.7814 ± 0.0018	
	z.landscape.h1	0.0046 ± 0.0014	0.9954 ± 0.0013	0.0019 ± 0.0001	0.9981 ± 0.0001	0.2254 ± 0.0080	0.7851 ± 0.0072	
	z.laplacian.h1	0.0044 ± 0.0015	0.9957 ± 0.0014	0.0018 ± 0.0001	0.9982 ± 0.0001	0.2326 ± 0.0160	0.7783 ± 0.0149	
	z.perslay.both	0.0046 ± 0.0014	0.9955 ± 0.0013	0.0012 ± 0.0002	0.9988 ± 0.0002	0.2398 ± 0.0158	0.7713 ± 0.0148	
	z.perslay.h0	0.0040 ± 0.0013	0.9961 ± 0.0012	0.0010 ± 0.0001	0.9990 ± 0.0001	0.2316 ± 0.0081	0.7791 ± 0.0075	
	z.perslay.h1	0.0044 ± 0.0014	0.9957 ± 0.0013	0.0018 ± 0.				

Table 15 – continued

Dataset	Mode	AE		GeoAE		TopoAE		
		MSE	R ²	MSE	R ²	MSE	R ²	
	z.diffusion_h1	0.0046 ± 0.0017	0.9955 ± 0.0016	0.0009 ± 0.0001	0.9991 ± 0.0001	0.2918 ± 0.0102	0.7192 ± 0.0096	
	z.fuse.diffusion_h1	0.0087 ± 0.0026	0.9914 ± 0.0025	0.0027 ± 0.0006	0.9973 ± 0.0006	0.3043 ± 0.0071	0.7071 ± 0.0066	
	z.fuse_h1	0.0087 ± 0.0028	0.9915 ± 0.0028	0.0027 ± 0.0005	0.9973 ± 0.0005	0.3037 ± 0.0119	0.7077 ± 0.0111	
	z.fuse.kpca_h1	0.0086 ± 0.0028	0.9916 ± 0.0027	0.0021 ± 0.0005	0.9979 ± 0.0005	0.3041 ± 0.0104	0.7073 ± 0.0097	
	z.fuse.landscape_h1	0.0080 ± 0.0030	0.9922 ± 0.0029	0.0026 ± 0.0005	0.9974 ± 0.0005	0.3126 ± 0.0146	0.6992 ± 0.0139	
	z.fuse.laplacian_h1	0.0084 ± 0.0033	0.9917 ± 0.0033	0.0018 ± 0.0007	0.9982 ± 0.0007	0.3043 ± 0.0125	0.7071 ± 0.0117	
	z.fuse.pca_h1	0.0081 ± 0.0033	0.9921 ± 0.0032	0.0015 ± 0.0005	0.9985 ± 0.0005	0.3000 ± 0.0098	0.7113 ± 0.0094	
	z.fuse.perslay_h1	0.0081 ± 0.0031	0.9921 ± 0.0030	0.0023 ± 0.0005	0.9977 ± 0.0005	0.3103 ± 0.0117	0.7013 ± 0.0110	
	z.fuse.pi_h1	0.0085 ± 0.0033	0.9916 ± 0.0032	0.0027 ± 0.0004	0.9973 ± 0.0004	0.3057 ± 0.0131	0.7058 ± 0.0124	
	z.fuse.rff_h1	0.0086 ± 0.0030	0.9916 ± 0.0029	0.0027 ± 0.0002	0.9973 ± 0.0002	0.3064 ± 0.0126	0.7052 ± 0.0119	
	z.h0	0.0048 ± 0.0017	0.9953 ± 0.0017	0.0009 ± 0.0001	0.9991 ± 0.0001	0.2881 ± 0.0066	0.7227 ± 0.0063	
	z.h1	0.0047 ± 0.0018	0.9954 ± 0.0017	0.0011 ± 0.0001	0.9989 ± 0.0001	0.2877 ± 0.0048	0.7231 ± 0.0043	
	z.landscape_both	0.0061 ± 0.0021	0.9940 ± 0.0020	0.0011 ± 0.0001	0.9989 ± 0.0001	0.2748 ± 0.0061	0.7355 ± 0.0058	
	z.landscape_h0	0.0052 ± 0.0018	0.9949 ± 0.0018	0.0010 ± 0.0002	0.9990 ± 0.0002	0.2812 ± 0.0091	0.7293 ± 0.0086	
	z.landscape_h1	0.0054 ± 0.0021	0.9947 ± 0.0020	0.0014 ± 0.0001	0.9986 ± 0.0001	0.2771 ± 0.0047	0.7333 ± 0.0044	
	z.laplacian_h1	0.0046 ± 0.0019	0.9954 ± 0.0019	0.0010 ± 0.0001	0.9990 ± 0.0001	0.2895 ± 0.0096	0.7214 ± 0.0090	
	z.perslay_both	0.0053 ± 0.0021	0.9948 ± 0.0020	0.0011 ± 0.0001	0.9989 ± 0.0001	0.2860 ± 0.0051	0.7247 ± 0.0046	
	z.perslay_h0	0.0048 ± 0.0017	0.9952 ± 0.0017	0.0012 ± 0.0001	0.9988 ± 0.0001	0.2817 ± 0.0100	0.7289 ± 0.0095	
	z.perslay_h1	0.0050 ± 0.0020	0.9951 ± 0.0019	0.0012 ± 0.0001	0.9988 ± 0.0001	0.2859 ± 0.0059	0.7248 ± 0.0054	
	z.pi_both	0.0060 ± 0.0021	0.9941 ± 0.0021	0.0014 ± 0.0002	0.9986 ± 0.0002	0.2930 ± 0.0089	0.7180 ± 0.0083	
	z.pi_h0	0.0059 ± 0.0022	0.9942 ± 0.0021	0.0015 ± 0.0003	0.9985 ± 0.0003	0.2954 ± 0.0110	0.7157 ± 0.0103	
	z.pi_h1	0.0048 ± 0.0016	0.9953 ± 0.0016	0.0013 ± 0.0003	0.9987 ± 0.0003	0.2855 ± 0.0069	0.7253 ± 0.0063	
	z.rff_h1	0.0048 ± 0.0020	0.9953 ± 0.0019	0.0011 ± 0.0002	0.9989 ± 0.0002	0.2903 ± 0.0056	0.7206 ± 0.0052	
	Lorenz-96	z	0.1536 ± 0.0057	0.8463 ± 0.0061	0.0012 ± 0.0003	0.9988 ± 0.0003	0.2726 ± 0.0298	0.7244 ± 0.0305
		topo_h1	0.2106 ± 0.0276	0.7892 ± 0.0281	0.0058 ± 0.0016	0.9942 ± 0.0016	0.5436 ± 0.1043	0.4503 ± 0.1062
		topo.pi_h1	0.2128 ± 0.0279	0.7870 ± 0.0284	0.0066 ± 0.0007	0.9934 ± 0.0007	0.5428 ± 0.1049	0.4510 ± 0.1067
z.both		0.2064 ± 0.0263	0.7935 ± 0.0267	0.0017 ± 0.0003	0.9983 ± 0.0003	0.2980 ± 0.0327	0.6987 ± 0.0334	
z.diffusion_h1		0.1505 ± 0.0031	0.8494 ± 0.0034	0.0012 ± 0.0002	0.9988 ± 0.0002	0.2726 ± 0.0309	0.7244 ± 0.0316	
z.fuse.diffusion_h1		0.2005 ± 0.0219	0.7993 ± 0.0223	0.0023 ± 0.0002	0.9977 ± 0.0002	0.3127 ± 0.0270	0.6838 ± 0.0276	
z.fuse_h1		0.1998 ± 0.0223	0.8000 ± 0.0227	0.0017 ± 0.0005	0.9983 ± 0.0005	0.3264 ± 0.0258	0.6699 ± 0.0265	
z.fuse.kpca_h1		0.2042 ± 0.0229	0.7956 ± 0.0233	0.0017 ± 0.0003	0.9983 ± 0.0003	0.3299 ± 0.0254	0.6664 ± 0.0260	
z.fuse.landscape_h1		0.2039 ± 0.0226	0.7959 ± 0.0230	0.0016 ± 0.0003	0.9984 ± 0.0003	0.3527 ± 0.0158	0.6433 ± 0.0163	
z.fuse.laplacian_h1		0.2031 ± 0.0285	0.7967 ± 0.0290	0.0018 ± 0.0007	0.9982 ± 0.0007	0.3263 ± 0.0222	0.6700 ± 0.0228	
z.fuse.pca_h1		0.2081 ± 0.0274	0.7917 ± 0.0278	0.0022 ± 0.0003	0.9978 ± 0.0003	0.4258 ± 0.0256	0.5695 ± 0.0260	
z.fuse.perslay_h1		0.2061 ± 0.0244	0.7937 ± 0.0248	0.0019 ± 0.0009	0.9981 ± 0.0009	0.3318 ± 0.0212	0.6645 ± 0.0217	
z.fuse.pi_h1		0.2058 ± 0.0275	0.7940 ± 0.0279	0.0013 ± 0.0003	0.9987 ± 0.0003	0.3446 ± 0.0175	0.6516 ± 0.0182	
z.fuse.rff_h1		0.1962 ± 0.0194	0.8036 ± 0.0198	0.0021 ± 0.0003	0.9979 ± 0.0003	0.3322 ± 0.0198	0.6641 ± 0.0203	
z.h0		0.2060 ± 0.0264	0.7938 ± 0.0269	0.0014 ± 0.0004	0.9986 ± 0.0004	0.2963 ± 0.0301	0.7004 ± 0.0307	
z.h1		0.1451 ± 0.0087	0.8548 ± 0.0089	0.0014 ± 0.0003	0.9986 ± 0.0003	0.2745 ± 0.0394	0.7224 ± 0.0402	
z.landscape_both		0.1838 ± 0.0256	0.8160 ± 0.0261	0.0018 ± 0.0003	0.9982 ± 0.0003	0.2873 ± 0.0335	0.7095 ± 0.0343	
z.landscape_h0		0.1662 ± 0.0058	0.8337 ± 0.0061	0.0018 ± 0.0005	0.9982 ± 0.0005	0.2826 ± 0.0352	0.7142 ± 0.0360	
z.landscape_h1		0.1456 ± 0.0091	0.8543 ± 0.0094	0.0018 ± 0.0002	0.9982 ± 0.0002	0.2785 ± 0.0336	0.7184 ± 0.0343	
z.laplacian_h1		0.1532 ± 0.0063	0.8467 ± 0.0065	0.0014 ± 0.0003	0.9986 ± 0.0003	0.2810 ± 0.0369	0.7158 ± 0.0376	
z.perslay_both		0.2076 ± 0.0272	0.7922 ± 0.0276	0.0012 ± 0.0001	0.9988 ± 0.0001	0.3664 ± 0.0302	0.6295 ± 0.0308	
z.perslay_h0		0.2074 ± 0.0274	0.7924 ± 0.0278	0.0013 ± 0.0003	0.9987 ± 0.0003	0.3512 ± 0.0255	0.6449 ± 0.0261	
z.perslay_h1		0.1437 ± 0.0127	0.8561 ± 0.0130	0.0013 ± 0.0003	0.9987 ± 0.0003	0.2783 ± 0.0397	0.7186 ± 0.0405	
z.pi_both		0.2037 ± 0.0224	0.7961 ± 0.0228	0.0015 ± 0.0003	0.9985 ± 0.0003	0.3233 ± 0.0338	0.6730 ± 0.0345	
z.pi_h0		0.2065 ± 0.0258	0.7933 ± 0.0263	0.0018 ± 0.0004	0.9982 ± 0.0004	0.3230 ± 0.0335	0.6735 ± 0.0341	
z.pi_h1		0.1420 ± 0.0119	0.8579 ± 0.0121	0.0016 ± 0.0002	0.9984 ± 0.0002	0.2761 ± 0.0311	0.7208 ± 0.0318	
z.rff_h1	0.1441 ± 0.0078	0.8558 ± 0.0080	0.0015 ± 0.0001	0.9985 ± 0.0001	0.2749 ± 0.0300	0.7220 ± 0.0307		
ElectricDevices	z	0.0139 ± 0.0072	0.9868 ± 0.0065	0.0014 ± 0.0001	0.9986 ± 0.0001	0.0615 ± 0.0047	0.9470 ± 0.0041	
	topo_h1	0.0927 ± 0.0167	0.9111 ± 0.0150	0.0081 ± 0.0016	0.9919 ± 0.0016	0.2020 ± 0.0271	0.8263 ± 0.0209	
	topo.pi_h1	0.0967 ± 0.0144	0.9072 ± 0.0127	0.0104 ± 0.0010	0.9896 ± 0.0010	0.2118 ± 0.0266	0.8178 ± 0.0202	
	z.both	0.0131 ± 0.0056	0.9876 ± 0.0050	0.0011 ± 0.0002	0.9989 ± 0.0002	0.0648 ± 0.0082	0.9440 ± 0.0078	
	z.diffusion_h1	0.0127 ± 0.0054	0.9880 ± 0.0047	0.0018 ± 0.0001	0.9982 ± 0.0001	0.0604 ± 0.0053	0.9479 ± 0.0053	
	z.fuse.diffusion_h1	0.0161 ± 0.0092	0.9848 ± 0.0083	0.0018 ± 0.0002	0.9982 ± 0.0002	0.1097 ± 0.0082	0.9055 ± 0.0070	
	z.fuse_h1	0.0167 ± 0.0094	0.9843 ± 0.0085	0.0017 ± 0.0001	0.9983 ± 0.0001	0.1210 ± 0.0144	0.8959 ± 0.0115	
	z.fuse.kpca_h1	0.0170 ± 0.0095	0.9839 ± 0.0085	0.0017 ± 0.0002	0.9983 ± 0.0002	0.1070 ± 0.0167	0.9078 ± 0.0138	
	z.fuse.landscape_h1	0.0175 ± 0.0100	0.9834 ± 0.0091	0.0022 ± 0.0002	0.9978 ± 0.0002	0.1415 ± 0.0114	0.8781 ± 0.0096	
	z.fuse.laplacian_h1	0.0172 ± 0.0100	0.9838 ± 0.0090	0.0018 ± 0.0005	0.9982 ± 0.0005	0.1169 ± 0.0081	0.8993 ± 0.0057	
	z.fuse.pca_h1	0.0172 ± 0.0106	0.9838 ± 0.0096	0.0014 ± 0.0002	0.9986 ± 0.0002	0.1205 ± 0.0105	0.8963 ± 0.0075	
	z.fuse.perslay_h1	0.0168 ± 0.0092	0.9842 ± 0.0084	0.0017 ± 0.0002	0.9983 ± 0.0002	0.1401 ± 0.0165	0.8793 ± 0.0141	
	z.fuse.pi_h1	0.0171 ± 0.0086	0.9839 ± 0.0077	0.0019 ± 0.0003	0.9981 ± 0.0003	0.1120 ± 0.0063	0.9034 ± 0.0065	
	z.fuse.rff_h1	0.0180 ± 0.0113	0.9830 ± 0.0103	0.0020 ± 0.0002	0.9980 ± 0.0002	0.1290 ± 0.0200	0.8891 ± 0.0161	
	z.h0	0.0133 ± 0.0053	0.9874 ± 0.0046	0.0012 ± 0.0003	0.9988 ± 0.0003	0.0713 ± 0.0082	0.9385 ± 0.0078	
	z.h1	0.0136 ± 0.0066	0.9871 ± 0.0059	0.0015 ± 0.0002	0.9985 ± 0.0002	0.0626 ± 0.0069	0.9460 ± 0.0065	
	z.landscape_both	0.0119 ± 0.0044	0.9887 ± 0.0039	0.0016 ± 0.0002	0.9984 ± 0.0002	0.0492 ± 0.0025	0.9576 ± 0.0028	
	z.landscape_h0	0.0117 ± 0.0053	0.9889 ± 0.0047	0.0014 ± 0.0001	0.9986 ± 0.0001	0.0532 ± 0.0027	0.9541 ± 0.0030	
	z.landscape_h1	0.0103 ± 0.0038	0.9902 ± 0.0033	0.0015 ± 0.0002	0.9985 ± 0.0002	0.0501 ± 0.0036	0.9568 ± 0.0036	
	z.laplacian_h1	0.0118 ± 0.0051	0.9888 ± 0.0045	0.0016 ± 0.0003	0.9984 ± 0.0003	0.0669 ± 0.0079	0.9424 ± 0.0070	
	z.perslay_both	0.0147 ± 0.0064	0.9861 ± 0.0057	0.0012 ± 0.0002	0.9988 ± 0.0002	0.0685 ± 0.0067	0.9410 ± 0.0063	
	z.perslay_h0	0.0127 ± 0.0050	0.9879 ± 0.0044	0.0020 ± 0.0005	0.9980 ± 0.0005	0.0670 ± 0.0067	0.9422 ± 0.0065	
	z.perslay_h1	0.0117 ± 0.0053	0.9889 ± 0.0047	0.0015 ± 0.0002	0.9985 ± 0.0002	0.0558 ± 0.0032	0.9519 ± 0.0034	
	z.pi_both	0.0155 ± 0.0062	0.9852 ± 0.0054	0.0038 ± 0.0003	0.9962 ± 0.0003	0.0746 ± 0.0094	0.9357 ± 0.0082	
	z.pi_h0	0.0166 ± 0.0057	0.9841 ± 0.0050	0.0035 ± 0.0006	0.9965 ± 0.0006	0.0688 ± 0.0106	0.9406 ± 0.0102	
	z.pi_h1	0.0102 ± 0.0031	0.9903 ± 0.0026	0.0015 ± 0.0003	0.9985 ± 0.0003	0.0579 ± 0.0052	0.9502 ± 0.0042	
z.rff_h1	0.0124 ± 0.0065	0.9883 ± 0.0059	0.0014 ± 0.0001	0.9986 ± 0.0001	0.0626 ± 0.0080	0.9461 ± 0.0070		
HeLa	z	0.0437 ± 0.0107	0.9429 ± 0.0151	0.0072 ± 0.0032	0.9928 ± 0.0031	0.0587 ± 0.0365	0.9205 ± 0.0576	
	topo_h1	0.0433 ± 0.0073	0.9435 ± 0.0109	0.0300 ± 0.0296	0.9693 ± 0.0305	0.0619 ± 0.0363	0.9164 ± 0.0583	
	topo.pi_h1	0.0816 ± 0.0654	0.8929 ± 0.0873	0.4645 ± 0.2120	0.5390 ± 0.2023	0.4991 ± 0.1520	0.3802 ± 0.1872	
	z.both	0.0413 ± 0.0069	0.9462 ± 0.0103	0.0079 ± 0.0031	0.9920 ± 0.0031	0.0602 ± 0.0359	0.9186 ± 0.0570	
	z.diffusion_h1	0.0409 ± 0.0051	0.9468 ± 0.0079	0.0087 ± 0.0039	0.9913 ± 0.0038	0.0586 ±		

Table 15 – continued

Dataset	Mode	AE		GeoAE		TopoAE	
		MSE	R^2	MSE	R^2	MSE	R^2
	z.fuse_rff_h1	0.0499 ± 0.0093	0.9349 ± 0.0136	0.0115 ± 0.0043	0.9885 ± 0.0041	0.0645 ± 0.0351	0.9135 ± 0.0566
	z.h0	0.0430 ± 0.0069	0.9440 ± 0.0103	0.0073 ± 0.0031	0.9926 ± 0.0030	0.0596 ± 0.0354	0.9196 ± 0.0561
	z.h1	0.0409 ± 0.0085	0.9466 ± 0.0121	0.0081 ± 0.0032	0.9919 ± 0.0030	0.0575 ± 0.0355	0.9222 ± 0.0565
	z.landscape_both	0.0432 ± 0.0074	0.9437 ± 0.0109	0.0073 ± 0.0033	0.9927 ± 0.0031	0.0610 ± 0.0364	0.9176 ± 0.0582
	z.landscape_h0	0.0440 ± 0.0077	0.9426 ± 0.0114	0.0073 ± 0.0035	0.9927 ± 0.0034	0.0623 ± 0.0380	0.9156 ± 0.0605
	z.landscape_h1	0.0383 ± 0.0083	0.9500 ± 0.0118	0.0071 ± 0.0033	0.9928 ± 0.0032	0.0550 ± 0.0324	0.9259 ± 0.0517
	z.laplacian_h1	0.0426 ± 0.0084	0.9444 ± 0.0120	0.0079 ± 0.0038	0.9921 ± 0.0037	0.0558 ± 0.0326	0.9249 ± 0.0517
	z.perslay_both	0.0437 ± 0.0086	0.9429 ± 0.0125	0.0078 ± 0.0028	0.9922 ± 0.0028	0.0608 ± 0.0363	0.9178 ± 0.0579
	z.perslay_h0	0.0438 ± 0.0079	0.9428 ± 0.0116	0.0081 ± 0.0033	0.9919 ± 0.0032	0.0608 ± 0.0358	0.9179 ± 0.0572
	z.perslay_h1	0.0389 ± 0.0081	0.9492 ± 0.0115	0.0076 ± 0.0033	0.9924 ± 0.0033	0.0572 ± 0.0329	0.9232 ± 0.0521
	z.pi_both	0.0507 ± 0.0082	0.9339 ± 0.0121	0.0319 ± 0.0228	0.9676 ± 0.0236	0.0804 ± 0.0482	0.8952 ± 0.0671
	z.pi_h0	0.0555 ± 0.0102	0.9276 ± 0.0147	0.0306 ± 0.0218	0.9688 ± 0.0226	0.0932 ± 0.0602	0.8811 ± 0.0746
	z.pi_h1	0.0405 ± 0.0046	0.9474 ± 0.0069	0.0159 ± 0.0087	0.9841 ± 0.0082	0.0607 ± 0.0321	0.9190 ± 0.0519
	z.rff_h1	0.0418 ± 0.0088	0.9455 ± 0.0126	0.0080 ± 0.0033	0.9920 ± 0.0032	0.0605 ± 0.0361	0.9182 ± 0.0577
Glioblastoma	z	0.0027 ± 0.0010	0.9970 ± 0.0011	0.0011 ± 0.0001	0.9988 ± 0.0001	0.0040 ± 0.0013	0.9957 ± 0.0015
	topo_h1	0.0100 ± 0.0125	0.9891 ± 0.0137	0.0730 ± 0.0965	0.9262 ± 0.0976	0.9931 ± 1.0330	−0.0730 ± 1.1188
	topo_pi_h1	0.0623 ± 0.0995	0.9345 ± 0.1036	0.1926 ± 0.1593	0.8058 ± 0.1605	0.2179 ± 0.1822	0.7660 ± 0.1966
	z.both	0.0030 ± 0.0009	0.9968 ± 0.0010	0.0013 ± 0.0005	0.9987 ± 0.0005	0.0035 ± 0.0003	0.9962 ± 0.0003
	z.diffusion_h1	0.0025 ± 0.0009	0.9973 ± 0.0010	0.0015 ± 0.0004	0.9985 ± 0.0004	0.0037 ± 0.0011	0.9959 ± 0.0012
	z.fuse_diffusion_h1	0.0060 ± 0.0016	0.9935 ± 0.0018	0.0079 ± 0.0019	0.9920 ± 0.0020	0.0104 ± 0.0041	0.9887 ± 0.0044
	z.fuse_h1	0.0100 ± 0.0024	0.9892 ± 0.0025	0.0063 ± 0.0035	0.9936 ± 0.0035	0.0078 ± 0.0015	0.9916 ± 0.0017
	z.fuse_kpca_h1	0.0086 ± 0.0022	0.9907 ± 0.0024	0.0062 ± 0.0019	0.9938 ± 0.0019	0.0081 ± 0.0017	0.9912 ± 0.0020
	z.fuse_landscape_h1	0.0107 ± 0.0033	0.9884 ± 0.0037	0.0076 ± 0.0028	0.9924 ± 0.0029	0.0089 ± 0.0021	0.9903 ± 0.0024
	z.fuse_laplacian_h1	0.0079 ± 0.0023	0.9915 ± 0.0026	0.0057 ± 0.0012	0.9943 ± 0.0012	0.0078 ± 0.0019	0.9915 ± 0.0022
	z.fuse_pca_h1	0.0079 ± 0.0025	0.9914 ± 0.0026	0.0066 ± 0.0029	0.9934 ± 0.0029	0.0092 ± 0.0031	0.9901 ± 0.0035
	z.fuse_perslay_h1	0.0071 ± 0.0016	0.9923 ± 0.0018	0.0075 ± 0.0021	0.9925 ± 0.0022	0.0096 ± 0.0034	0.9895 ± 0.0037
	z.fuse_pi_h1	0.0104 ± 0.0051	0.9888 ± 0.0055	0.0209 ± 0.0217	0.9789 ± 0.0219	0.0100 ± 0.0016	0.9891 ± 0.0019
	z.fuse_rff_h1	0.0071 ± 0.0019	0.9923 ± 0.0021	0.0080 ± 0.0019	0.9919 ± 0.0019	0.0075 ± 0.0012	0.9918 ± 0.0014
	z.h0	0.0026 ± 0.0010	0.9972 ± 0.0011	0.0015 ± 0.0002	0.9985 ± 0.0002	0.0038 ± 0.0011	0.9959 ± 0.0012
	z.h1	0.0026 ± 0.0005	0.9972 ± 0.0006	0.0013 ± 0.0005	0.9986 ± 0.0005	0.0040 ± 0.0008	0.9957 ± 0.0010
	z.landscape_both	0.0025 ± 0.0007	0.9972 ± 0.0008	0.0012 ± 0.0003	0.9988 ± 0.0003	0.0040 ± 0.0012	0.9957 ± 0.0014
	z.landscape_h0	0.0026 ± 0.0010	0.9972 ± 0.0011	0.0008 ± 0.0001	0.9992 ± 0.0001	0.0038 ± 0.0012	0.9958 ± 0.0014
	z.landscape_h1	0.0026 ± 0.0002	0.9972 ± 0.0003	0.0012 ± 0.0003	0.9988 ± 0.0003	0.0037 ± 0.0011	0.9960 ± 0.0012
	z.laplacian_h1	0.0032 ± 0.0012	0.9965 ± 0.0013	0.0012 ± 0.0005	0.9988 ± 0.0005	0.0040 ± 0.0015	0.9957 ± 0.0017
	z.perslay_both	0.0035 ± 0.0002	0.9962 ± 0.0003	0.0012 ± 0.0003	0.9988 ± 0.0003	0.0040 ± 0.0010	0.9956 ± 0.0012
	z.perslay_h0	0.0038 ± 0.0010	0.9959 ± 0.0011	0.0015 ± 0.0003	0.9985 ± 0.0003	0.0043 ± 0.0009	0.9953 ± 0.0010
	z.perslay_h1	0.0021 ± 0.0006	0.9977 ± 0.0006	0.0015 ± 0.0006	0.9985 ± 0.0006	0.0033 ± 0.0008	0.9964 ± 0.0010
	z.pi_both	0.0072 ± 0.0027	0.9923 ± 0.0028	0.0056 ± 0.0023	0.9944 ± 0.0023	0.0087 ± 0.0020	0.9906 ± 0.0023
	z.pi_h0	0.0064 ± 0.0023	0.9931 ± 0.0023	0.0059 ± 0.0042	0.9941 ± 0.0042	0.0062 ± 0.0013	0.9933 ± 0.0015
	z.pi_h1	0.0033 ± 0.0011	0.9965 ± 0.0011	0.0034 ± 0.0027	0.9966 ± 0.0027	0.0038 ± 0.0007	0.9959 ± 0.0009
	z.rff_h1	0.0022 ± 0.0005	0.9976 ± 0.0005	0.0017 ± 0.0005	0.9983 ± 0.0005	0.0048 ± 0.0015	0.9948 ± 0.0017

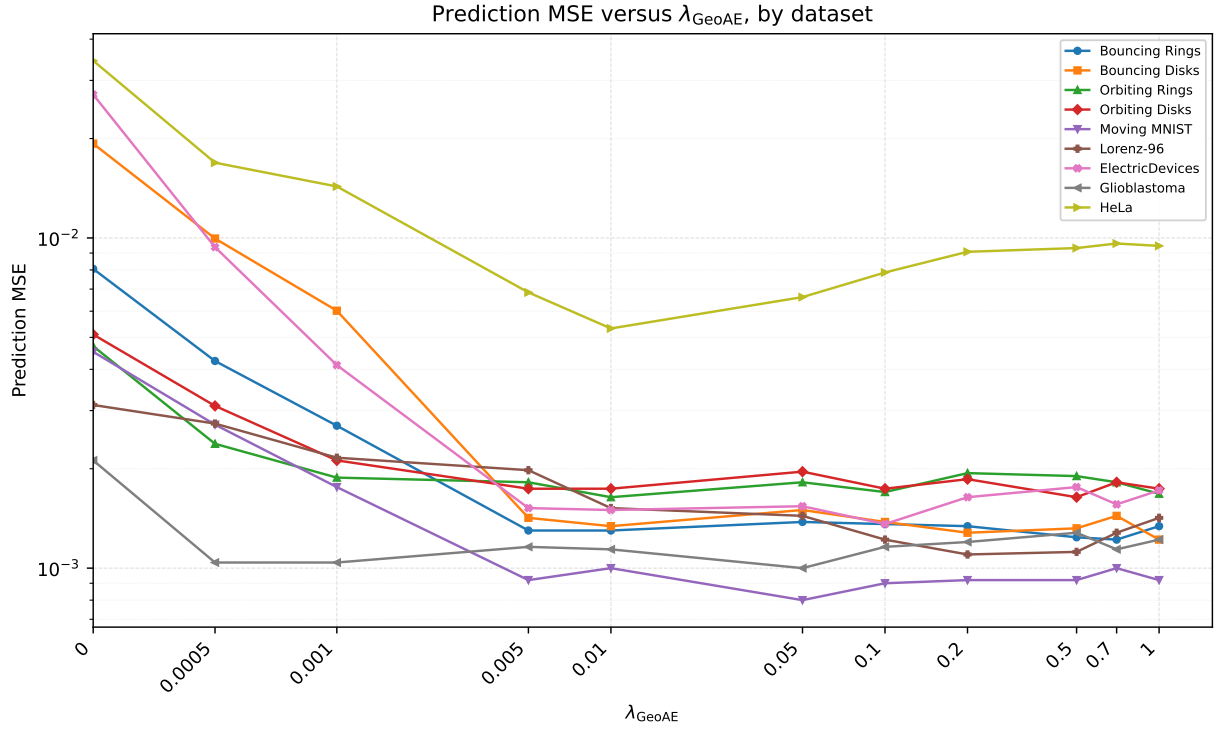


Figure 10: Prediction MSE vs λ_{GeoAE} , per dataset

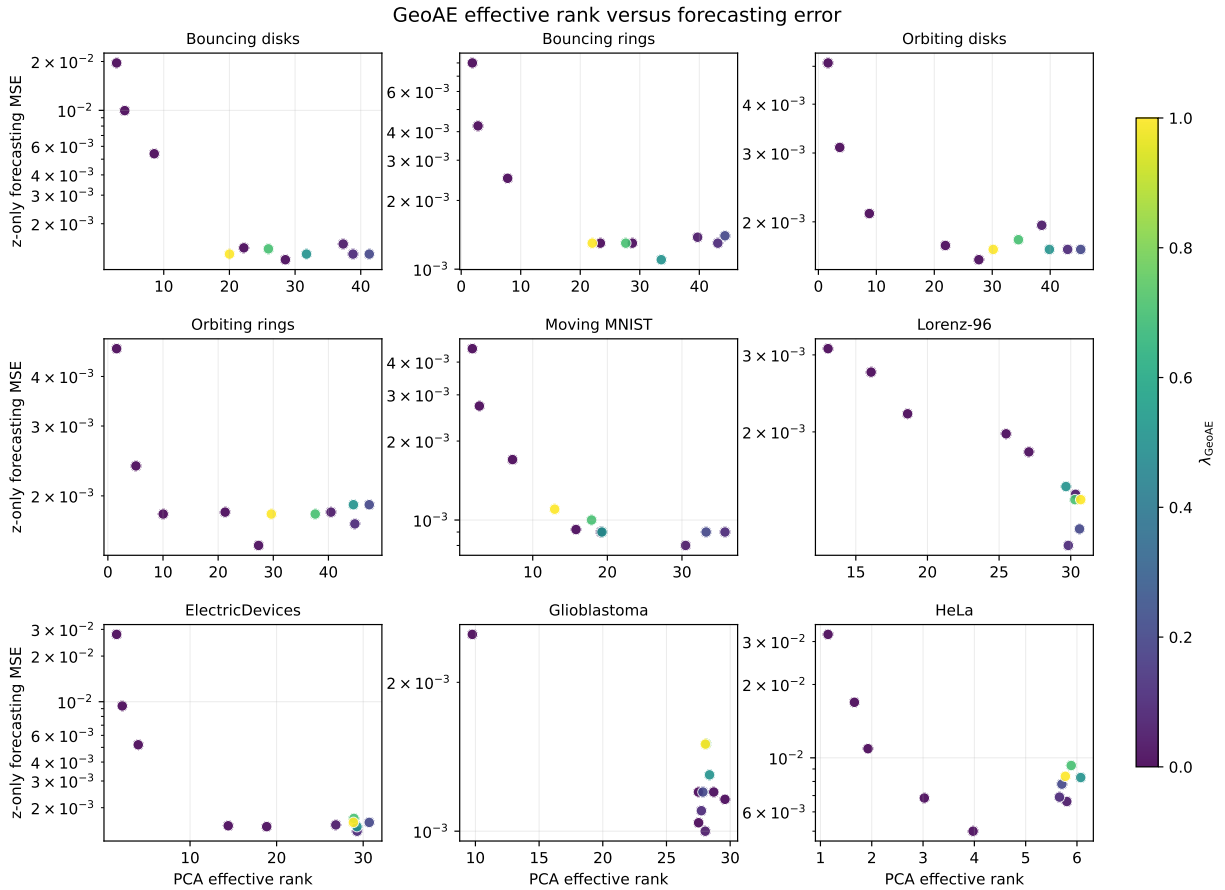


Figure 11: z -only prediction MSE vs PCA rank, per dataset

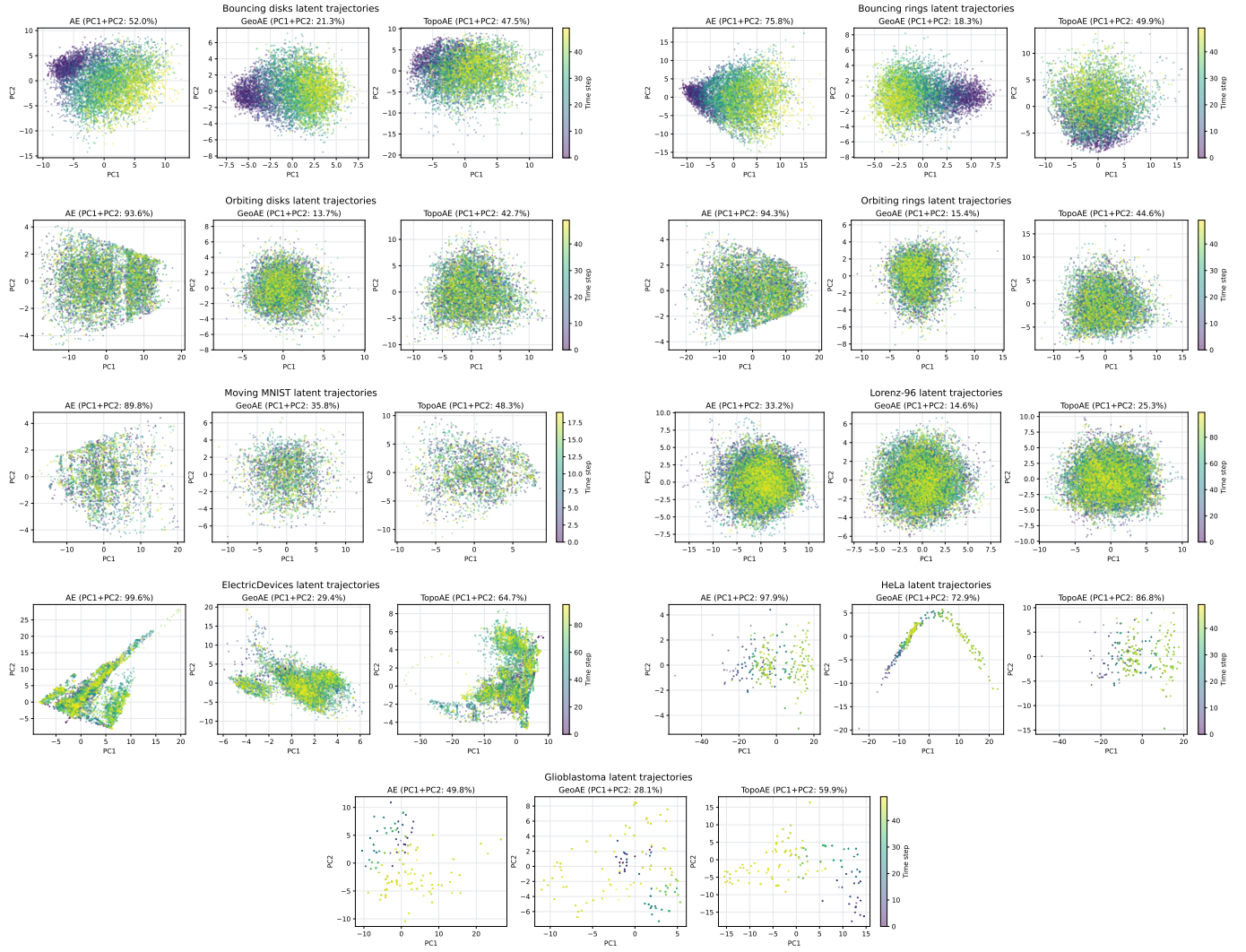


Figure 12: PCA projections for each dataset and encoder variant. Each panel header reports the percentage of total latent variance explained jointly by PC1 and PC2.